

UNIVERSITY OF ZAGREB
FACULTY OF ELECTRICAL ENGINEERING AND COMPUTING

MASTER'S THESIS No. 1337

**CLASSIFICATION OF AFFECTIVE DISOR-
DERS FROM EEG SIGNALS USING POWER
SPECTRUM AND CNN**

Lucija Čorak

Zagreb, June 2026

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MASTER'S THESIS ASSIGNMENT No. 1337

Candidate: **Lucija Čorak (0036532673)**

Study programme: Computing

Profile: Data Science

Supervisor: Prof. Mario Cifrek, PhD

Assignment: **Classification of Affective Disorders from EEG Signals Using Power Spectrum and CNN**

Description of the assignment:

Diagnosing affective disorders is challenging because of the subjectivity of symptoms and the lack of objective biomarkers. One direction in the classification of affective disorders focuses on the use of deep learning models in which the EEG signal is used as the input. The convolutional neural network (CNN) is an example of a deep learning model that is extremely effective in the analysis of visual data, such as 2D spectrograms. In the thesis, it is necessary to study the literature on power spectrum extraction and on CNNs. Investigate in detail the application of EEG signals in the classification of affective disorders. Study the available EEG signal database with labelled diagnoses of affective disorders. Develop a procedure for extracting the power spectrum of EEG signals and transforming it into a 2D form. Implement and optimize a CNN model for the classification of affective disorders on the transformed data. Apply the Grad-CAM method for model explainability and visualization of relevant EEG features. Compare the obtained classification results, using basic evaluation measures (e.g. accuracy, F1-score), with results from the literature. For detailed information, contact Eda Jovičić, mag. ing.

Deadline for submission: 26 June 2026

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Abstract

Classification of Affective Disorders from EEG Signals Using Power Spectrum and CNN

Affective disorders are mental disorders characterized by abrupt changes in mood, and electroencephalography is a non-invasive method of measuring the electrical activity of the brain. The aim of this thesis was to analyze and compare the EEG signals of individuals diagnosed with an affective disorder and of healthy individuals. Classification was carried out on the basis of 2D representations of EEG signals, such as the power spectrum, as input information to a convolutional neural network (CNN).

Keywords: affective disorders, electroencephalography, EEG, power spectrum of EEG signals, deep learning, convolutional neural network

Summary

Classification of Affective Disorders for EEG Signals using power spectrum and CNN

Affective disorders are mental disorders which are characterized by sudden mood changes and electroencephalography is a non-invasive method of measuring brain electric activity. The goal of this paper was to analyze and compare EEG signals of individuals diagnosed with an affective disorder and healthy individuals. The classification was based on 2D depictions of EEG signals, such as power spectrum, which were used as input into a Convolutional Neural Network (CNN).

Keywords: affective disorders, electroencephalography, EEG, power spectrum of EEG signals, deep learning, Convolutional Neural Network (CNN)

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Introduction

Affective disorders are a type of mental disorder characterized by the appearance of pronounced, abrupt changes in an individual's emotions. They are divided into depressive disorder and bipolar disorder. These types of illness represent a significant problem for global healthcare because there is a large number of affected people, yet they are difficult to diagnose correctly. The reason for this is that the diagnostic procedure is based on a subjective analysis of symptoms and clinical signs and is specific to each individual. Given these problems, the application of various methods from the field of neuroscience, such as electroencephalography, is increasingly being investigated in order to develop objective diagnostic tools for affective disorders.

Electroencephalography (EEG) is a non-invasive method of measuring brain potentials that is increasingly used for studying and diagnosing various mental disorders, including affective disorders. Given the large number of people with these illnesses, ever more efficient and reliable ways of diagnosing affective disorders are needed in order to determine the correct diagnoses and suitable therapies for each affected person. From this arises the idea of applying artificial intelligence and deep learning to automate the diagnostic process and reduce the time doctors and experts need to manually review each individual's electroencephalograms.

The aim of this thesis is to investigate the possibilities of using various 2D representations of EEG signals for distinguishing the EEG signals of individuals with an affective disorder from those of healthy individuals. Then, on the obtained representations, to apply a convolutional neural network for the purpose of classification, to study the classification accuracy measures, and to study, using the Grad-CAM method, on the basis of which region of the image the model makes its decision.

The first part of the thesis explains electroencephalography and describes affective disorders in more detail. Next, the EEG dataset on which the thesis is based is presented, followed by the presentation of the 2D representations of EEG signals used later in the thesis. In the fifth chapter, the field of deep learning, convolutional neural networks and the Grad-CAM method are presented, and in the sixth chapter the obtained results are shown.

1. Electroencephalography

Understanding the basic neurological processes that underlie complex higher-order cognitive operations and functional domains is a fundamental goal of cognitive neuroscience. Electroencephalography (EEG) is a non-invasive and relatively inexpensive method for assessing neurophysiological function that can be used to achieve this goal. EEG measures the electrical activity of large, synchronously active populations of neurons in the brain with electrodes placed on the scalp [1]. Owing to its non-invasiveness and its cost-benefit ratio, EEG is the most commonly used brain recording method in clinical research, laboratory experiments, patient health monitoring, diagnostics and many other applications. Unfortunately, EEG recordings are often contaminated by various types of artifacts, such as artifacts caused by electrode movement, movement artifacts, ocular artifacts and EMG artifacts from muscle activity [2].

When recording the electrical potentials of the brain with an electroencephalograph, the electrodes are usually placed in a particular arrangement on the surface of the scalp, and the most common arrangement is the 10-20. Such an arrangement means placing electrodes at 10% or 20% of a previously measured distance between two points on the head (from ear to

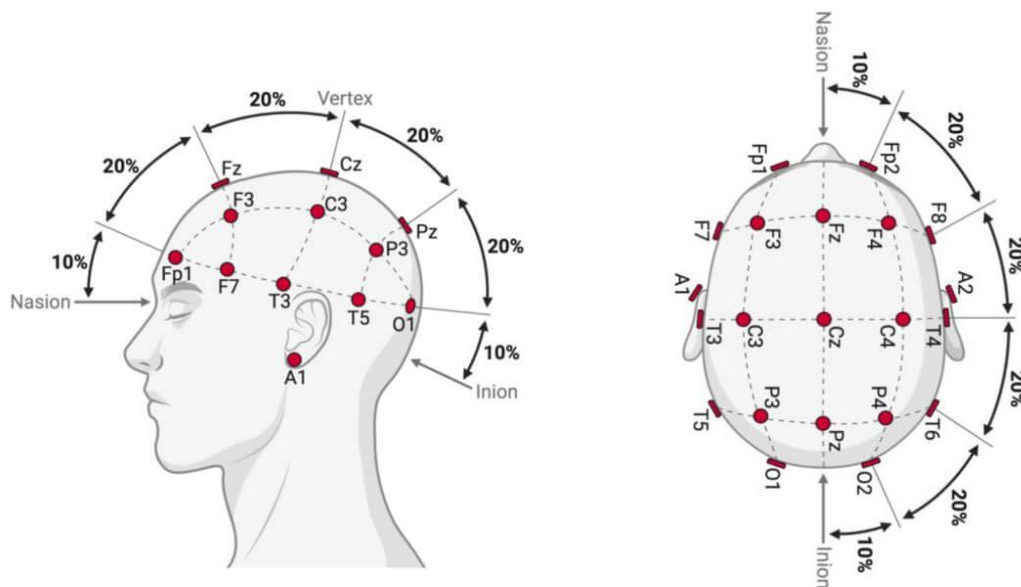


Figure 1.1. Electrode placement according to the 10-20 system [3]

ear and frominion to nasion), which is used in order to preserve the same distance of the electrodes across different subjects (Figure 1.1.).

The brain potential recorded by electrodes on the surface of the head is very weak, since the recorded signals travel from the surface of the brain through the cerebrospinal fluid, then the skull bone, skin tissue and subcutaneous fat, which act like a low-pass filter, so the recorded potential ends up with an amplitude of the order of magnitude of microvolts (μV) [4]. Before analysis and drawing conclusions on the basis of the recorded signals, it is necessary to process them and remove interference from them. Interference or artifacts in the signal can be body movement, electrode movement, heart activity, eye movement, interference from the electrodes used for recording, mains interference, etc.

On the obtained electroencephalogram, after the artifacts have been removed, five different types of brain waves can be observed. They are divided into α , β , γ , δ and θ waves, which appear in particular brain states and differ from one another in frequency and amplitude (Figure 1.2.).

1.1. Types of brain waves

Alpha waves have an amplitude between 10 and 45 μV and a frequency between 8 and 12 Hz, and they predominate in the occipital region of the brain in subjects who are awake and relaxed with their eyes closed.

When a subject remains awake and relaxed and opens their eyes, there is a considerable decrease in the presence of α waves, and the presence of β waves increases. Their amplitude is in the range of 10 to 20 μV and their frequency is in the range of 12 to 30 Hz. This type of wave also appears during mental activity, and is most often observed in the frontal and parietal regions of the brain.

Gamma waves appear during concentration and use of memory, and may also appear spontaneously during the REM phase of sleep (*Rapid Eye Movement*). They are the most difficult to study because their frequency varies from 30 to 100 Hz, and artifacts caused by external interference also occur in this range, so they are difficult to isolate.

Delta waves denote low-frequency activity in the range of 1 to 4 Hz that appears in the cerebral cortex and that is most prevalent in deep sleep, in newborns and in severe brain diseases.

Theta waves usually appear in states of drowsiness, meditation and in some stages of sleep, specifically those relating to “light” sleep [5].

Type and Range	Waveform	Function
Gamma wave more than 30 Hz		Concentration, focus, and learning
Beta wave 13 - 30 Hz		Most activities in a waking state
Alpha wave 8 - 13 Hz		Dreaming, broad consciousness
Theta wave 4 - 8 Hz		Relaxation, dreamy state
Delta wave 1 - 4 Hz		Deep sleep

Figure 1.2. Types of brain waves [6]

2. Affective disorders

Affective disorders are types of mental disorders characterized by abrupt changes in an individual's emotions [7], and they may contain depressive or manic episodes or an alternation of both. The World Health Organization estimated in 2019 that 280 million people in the world suffer from depressive disorder and 40 million people suffer from bipolar disorder, and that a large number of people still remain without a diagnosis or with an incorrect one. These disorders affect the everyday functioning of individuals, have a strong impact on social interactions and can significantly reduce quality of life. The most general division of affective disorders is into depressive and bipolar disorders.

People with depressive disorder experience depressive episodes, during which the person experiences a depressed mood (feelings of sadness, irritability, emptiness) or a loss of pleasure or interest in any activities, for most of the day, nearly every day, for a period of at least two weeks. Other symptoms are also present, which may include poor concentration, feelings of excessive guilt or low self-worth, hopelessness about the future, thoughts of dying or suicide, disturbed sleep, changes in appetite or body weight, and a feeling of particular exhaustion or low energy. People with depression are at increased risk of suicide. Nevertheless, effective psychological therapies exist and, depending on age and severity, drug therapy is also considered [8].

People with bipolar disorder experience alternating depressive episodes with periods of manic symptoms. During a depressive episode, the person experiences a depressed mood (feeling of sadness, irritability, emptiness) or a loss of pleasure or interest in activities, for most of the day and nearly every day. Manic symptoms may include euphoria or irritability, increased activity or energy, and other symptoms such as increased talkativeness, racing thoughts, increased self-confidence, reduced need for sleep, distractibility, and impulsive, risky behavior. People with bipolar disorder are at increased risk of suicide. Nevertheless, effective treatment options exist, including psychological education, stress reduction and strengthening of social functioning, as well as drug therapy [8].

For the purposes of this thesis, the classification of affective disorders issued by the World Health Organization will be used, which is called the International Classification of Diseases and Related Health Problems (*International Classification of Diseases – Tenth*

Edition, ICD-10). In ICD-10, diseases and health problems are systematically divided into groups according to similarities in etiology, symptoms and course of the disease, and each has its own unique numerical code that facilitates unambiguous identification and classification. The levels for describing the degree of an affective disorder are divided into mild, moderate and severe, and there is also a division into single episodes and recurrent disorders (a larger number of episodes). Diagnoses are made by the assessment of doctors who have access to certain questionnaires, by means of which the typical symptoms of a particular affective disorder, defined according to the ICD-10 manual, can be established. The severity of the illness is then determined according to the amount of inability to carry out everyday physical, mental and work obligations and according to the duration of the symptoms.

2.1. Depressive episode and recurrent depressive disorder

Since the dataset used contains recordings of people with depressive episodes (F32) and people with recurrent depressive disorder (F33), only these will be considered from here on.

In typical mild, moderate or severe depressive episodes the patient has a lowered mood, less energy and reduced activity. The capacity for enjoyment, interest and concentration are reduced, and fatigue is often pronounced even after the slightest effort. Sleep is usually disturbed and appetite reduced. Self-esteem and self-confidence are almost always reduced, and even in the milder form some ideas of guilt and worthlessness are often present. The lowered mood varies little from day to day, is not influenced by circumstances, and may be accompanied by so-called somatic symptoms, such as loss of interest and feelings of pleasure, waking in the morning several hours before the usual time, depression being worse in the morning, marked psychomotor retardation, agitation, loss of appetite, loss of body weight and loss of libido. Depending on the number and severity of the symptoms, a depressive episode can be designated as mild, moderate or severe [9].

Depressive episodes can be divided by symptom severity into:

1. F32.0 - Mild depressive episode
2. F32.1 - Moderate depressive episode
3. F32.2 - Severe depressive episode without psychotic symptoms
4. F32.3 - Severe depressive episode with psychotic symptoms

5. F32.8 - Other depressive episodes
6. F32.9 - Depressive episode, unspecified

Recurrent depressive disorder is a type of affective disorder that is listed in the ICD-10 system under the code F33. The characteristics of this disorder are recurring episodes of depression without the occurrence of manic episodes, and the individual shows reduced energy and activity and feels dejected, sad and listless. The consequences of this disorder can be sleep disturbances, changes in appetite and body mass, and in the worst cases thoughts of self-harm or of committing suicide. The duration of such episodes in patients can vary, from several weeks to sometimes even several months. The occurrence of psychotic symptoms is also possible in more severe depressive episodes, and these include hallucinations, delusions or some other forms of loss of contact with reality.

There are several different types of recurrent depressive disorder, which are designated as follows:

1. F33.0 - recurrent depressive disorder, current episode mild
2. F33.1 - recurrent depressive disorder, current episode moderate
3. F33.2 - recurrent depressive disorder, current episode severe, without psychotic symptoms
4. F33.3 - recurrent depressive disorder, current episode severe, with psychotic symptoms
5. F33.4 - recurrent depressive disorder, currently in remission
6. F33.8 - other recurrent depressive disorders
7. F33.9 - recurrent depressive disorder, unspecified

3. EEG signal dataset

3.1. Recording of the dataset

For the purposes of this project, a database recorded at the Vrapče Psychiatric Hospital in Zagreb was used. The EEG signals were recorded from healthy persons who voluntarily agreed to take part in the research and from patients at the clinic as part of the standard examination process, which is practice in order to check for the possible presence of brain damage or other physical causes of the occurrence of affective disorders.

The dataset consists of 70 healthy “control” subjects and 70 subjects diagnosed with some type of depressive episode or recurrent depressive disorder. Originally, in the period between 2016 and 2020, 105 recordings were made, of which 35 were of healthy subjects and 70 of affected subjects. In order to equalize the number of positive and negative examples in the classes for classification problems, a further 35 recordings of healthy subjects were subsequently made in 2024. In order to preserve the anonymity of the subjects, they were assigned different four-digit identification numbers within the research or, in the later recordings, a code consisting of a combination of 8 letters and digits. For each recorded EEG signal, the table records the subject's identification number, the date of the recording, the duration of the recording, the number of recorded signal samples and the sampling frequency. In addition to the stated data on the EEG signals, the following data on the subjects were also recorded: sex, year of birth, diagnosis, presence of a personality disorder, number of therapies, date of onset of the disorder, date of onset of the last episode, rehospitalization, use of medication (benzodiazepine, stabilizers, antipsychotics, antidepressants, type of antidepressant) and the subject's state during the recording (mood, dynamism, energy, appetite, sleep, cognitive abilities, auto-aggression, pessimism and insight into their condition) [10].

During the recording of the EEG, the subjects were in a calm and quiet room in a relaxed state. First, data about the subject were collected, and following that the recording procedure itself was described to them. The subject was then asked to sit down so that a cap with electrodes for recording the electroencephalogram could be placed on their head. A cap with electrodes placed in the 10-20 arrangement (described in Chapter 1). It carried 19 EEG electrodes and 11 other channels: electromyographic (EMG), photoelectronystagmographic (PNG), electrocardiographic (ECG) and event marker channels (MKR). In the first

recording, Oz was used as the reference electrode, while the measuring electrodes were Fp1, Fp2, F7, F3, Fz, F4, F8, T3, C3, Cz, T4, T5, P3, Pz, P4, T6, O1 and O2 [10] (Figure 3.1.). In the second recording the electrode arrangement remained the same, but they were named differently, so the reference electrode is Oz-Ref, and the measuring electrodes are Fp1-Ref, Fp2-Ref, F7-Ref, F3-Ref, Fz-Ref, F4-Ref, F8-Ref, T3-Ref, C3-Ref, Cz-Ref, T4-Ref, T5-Ref, P3-Ref, Pz-Ref, P4-Ref, T6-Ref, O1-Ref and O2-Ref (Figure 3.2.). The sampling frequency of the signal differs depending on the year of recording, so the older device used in the first recording had a sampling frequency of 256 Hz, and the newer device, used in 2024, had a sampling frequency of 200 Hz. Because the data need to have the same sampling frequency, the older dataset was resampled to a frequency of 200 Hz. [10]

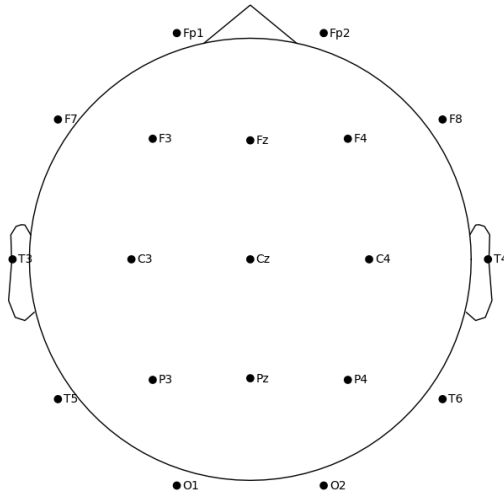


Figure 3.1.1. Electrode positions of the first recording

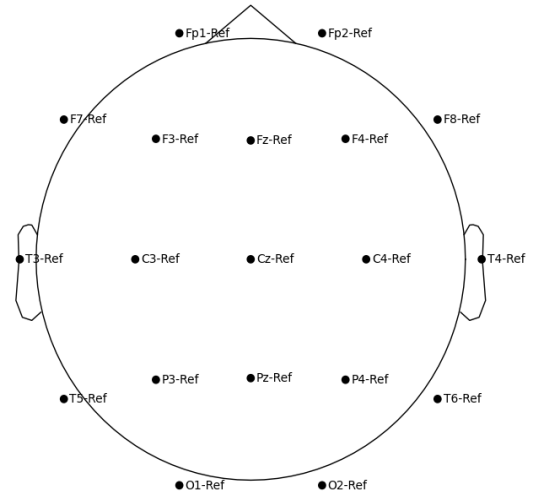


Figure 3.1.2. Electrode positions of the second recording

During the recording process itself, the subject first lay down so as to be able to relax completely and so as to minimize the use of muscles and eliminate as much as possible other biological sources of artifacts that could otherwise affect the recorded data. The recording of the EEG signal for an individual subject lasted about 30 minutes. During the examination, the subject remained still and alternately opened their eyes and kept them open for 8 to 10 seconds, and after that closed their eyes and kept them closed for approximately 45 seconds. This process was repeated 5 to 7 times, and after it photostimulation was performed with five different flashing frequencies, at the frequencies of: 4 Hz, 8 Hz, 16 Hz, 24 Hz and 30 Hz. The last step of the examination was the induction of hyperventilation of the subject,

according to the instructions of the recording technician. The recording process is shown in Figure 3.1.3.

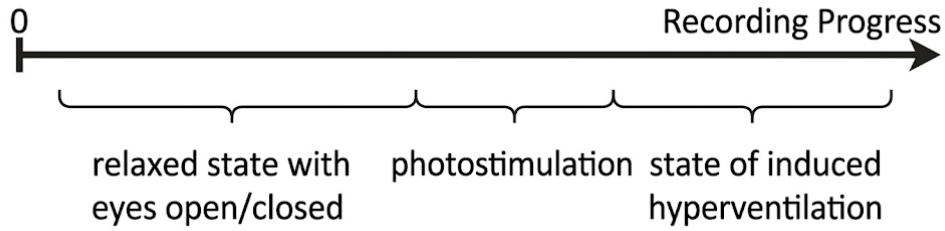


Figure 3.1.3. Recording process [11]

3.2. Preprocessing of the dataset

After the recording of the EEG signal, the next important step is signal preprocessing, because the recorded signals consist of the brain signal and various noises or artifacts. Their source may be, for example: heart activity, movement of muscles or eyes, electrode noise, mains frequency or the operating frequency of surrounding devices present during recording. In general, the process of EEG signal preprocessing consists of filtering (removal of interference), averaging the signal to a common reference, and independent component analysis (ICA). This step is necessary because all recorded potentials whose source was not brain activity are considered interference in the EEG [11], and they need to be removed in order to obtain a signal that represents true brain activity.

The dataset used in this thesis was, after recording, loaded and preprocessed in the MATLAB environment using the tool *eeglab*. The first step was to equalize the sampling frequency of the entire dataset, so the sampling frequency of the first, older part of the dataset was reduced to 200 Hz. In order to remove low-frequency and high-frequency noise from the EEG signal, a band-pass finite impulse response (*FIR*) filter, shown in Figure 3.2.1., was then applied to all examples, in the range from 0.1 Hz to 40 Hz. This removed low-frequency interference in the signal whose source is head movements during recording, movements of electrode wires and sweating of the scalp, as well as high-frequency interference such as mains noise, muscle contractions and electromagnetic interference.

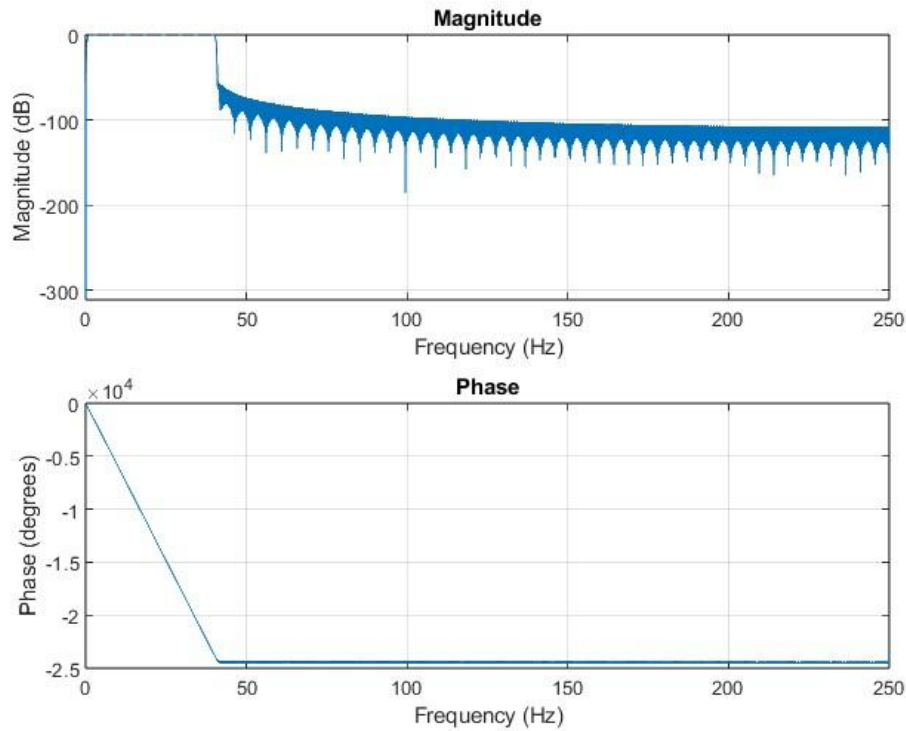


Figure 3.2.1. Amplitude and phase response of the Basic FIR filter in EEGLAB

After filtering, the channels were re-referenced to the average value of all 19 channels [10]. In this way each EEG signal channel is adjusted to the reference channel, which is calculated as the mean value of all channels. In this way the focus of the signal analysis is redirected to the relative changes in activity between different channels, instead of to the absolute activity with respect to the reference channel, which makes it possible to investigate the interrelations of various parts of the brain.

As the last step of the analysis of the EEG signal, independent component analysis (*ICA*) is applied. This method is generally used for extracting individual components from a set of signals. With EEG signals, ICA serves to identify and extract individual components that correspond to the activities of specific brain regions. However, a drawback of this method is the assumption that the number of observed signal mixtures must be equal to the number of signal sources [10]. On the dataset used, by means of the semi-automated *eeglab* classifier ICLabel, it was estimated (with a certain level of confidence) what the source of a given component in the signal is (Figure 3.2.2.). The algorithm divided the 19 components into different categories with regard to their source: brain, eyes, heart, muscles, signal noise and other. On the basis of their categorization, the components considered to be artifacts were

reviewed and manually removed from the signal, and in this way the preprocessed EEG signal used further in this thesis was obtained.

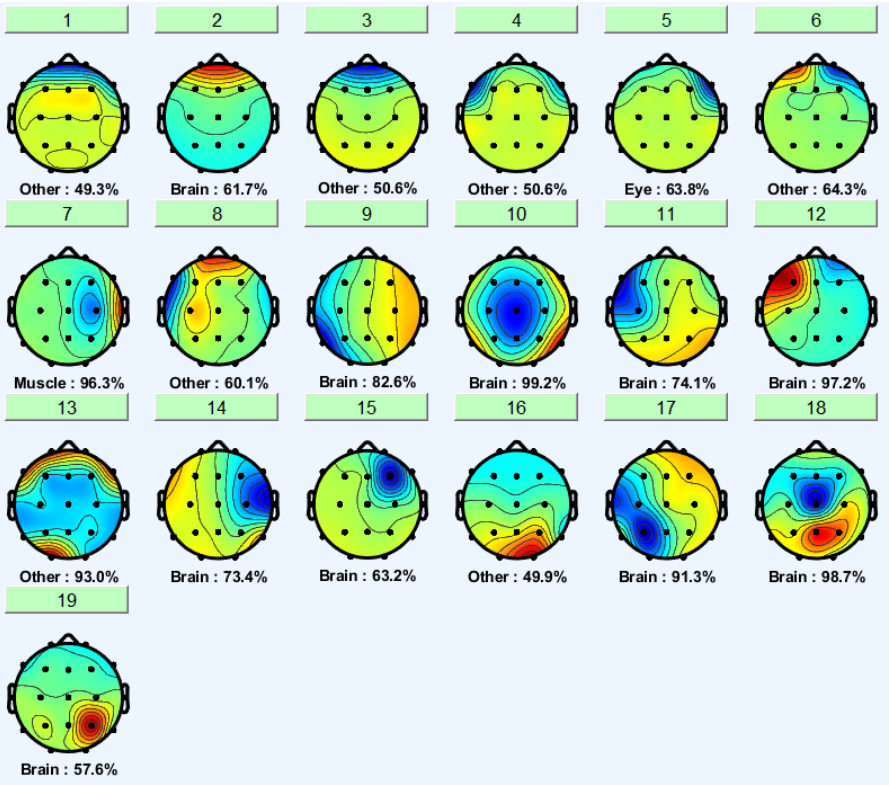


Figure 3.2.2. Signal decomposed into components and classified by the semi-automated algorithm ICLabel

4. 2D representations of EEG signals

In the analysis of EEG signals, data visualization plays a key role in understanding their structure, spatial distribution, interdependence and dynamics. The original signals consist of multichannel time series of high temporal resolution, so their interpretation in the original form is difficult, especially when it comes to longer recordings, a large number of subjects or different mental states. For this reason, computational analysis of EEG signals has been intensively developed in recent decades, in order to facilitate understanding and the extraction of information for further interpretation. Analysis of signals in the context of identifying patterns that can be associated with cognitive and emotional states is increasingly being considered, which is at the same time important in the field of research on affective disorders.

Discovering the mentioned patterns is a very demanding task. EEG signals are recorded with surface electrodes and may contain noise caused by electrical interference or by movement of electrodes on the scalp. An additional problem is posed by eye blinks and other muscle activity, which can produce signals of greater amplitude than those originating from brain activity. In addition, there are also problems of a cognitive nature, since the subject's level of concentration can vary even while performing the same mental task [12]. For this reason, various methods of visualization and transformation of signals are used, with 2D representations playing an important role in highlighting spatial and temporal patterns of activity. The mentioned representations enable easier analysis of differences in EEG signals that may be associated with emotional and affective states and represent an important step in the further analysis and modelling of the data.

For the purposes of this thesis, 2D representations of the spectral power of EEG signals, spectrograms of EEG signals and topographic maps of EEG signals will be considered.

4.1. Power spectrum of EEG signals

Spectral analysis is one of the standard methods used for the quantification of EEG signals. Power spectral density (PSD), i.e. the power spectrum, reflects the “frequency content” of a signal or the distribution of the signal's power over frequencies [13] (Figure 4.1.1.).

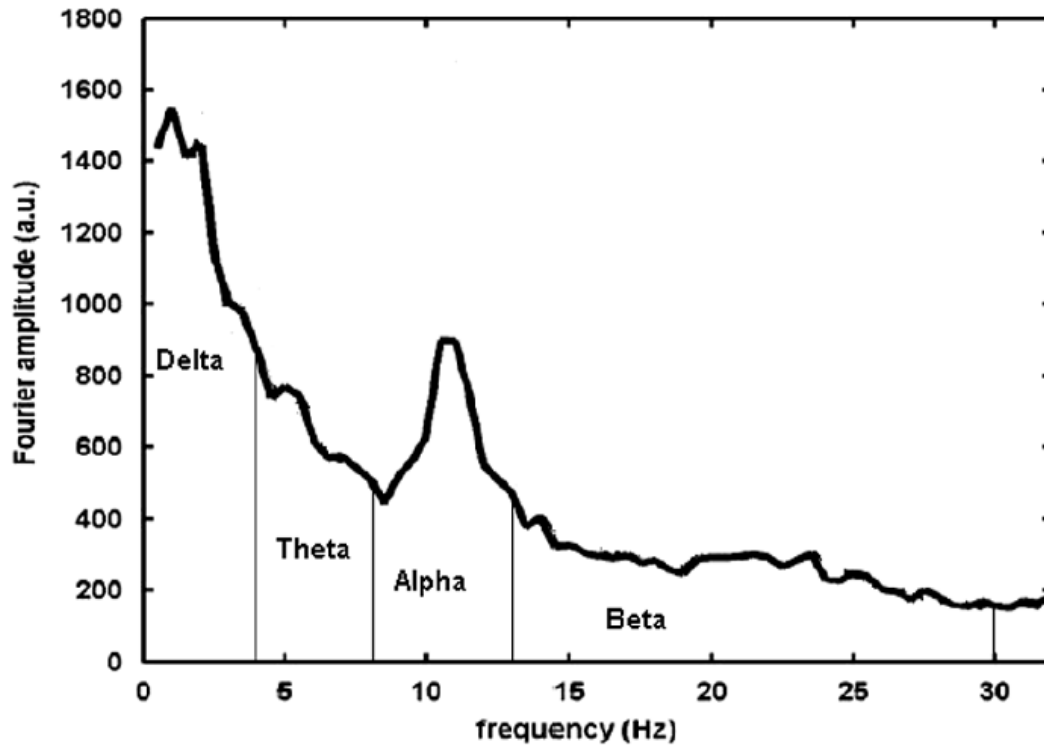


Figure 4.1.1. Spectral power distribution of a normal human EEG at rest [14]

Namely, by identifying changes in neuronal oscillations associated with stress, PSD provides valuable information for assessing and monitoring stress levels. This method is relatively simple to implement, requires fewer computational resources compared with more complex techniques, and because of its reliability and practicality it is widely used in scientific research and publications. To apply the power spectral density method, it is necessary to calculate the power of the signal within a certain frequency range. There are two ways: the fast Fourier transform (FFT) or Welch's method. Welch's method is often preferred because of its greater accuracy and reliability in estimating power spectral density, while FFT is desirable because of its computational efficiency and speed of execution.

For the purposes of this thesis, Welch's method was used, which makes it possible to reduce the influence of noise in the estimated spectrum. The procedure of Welch's method is based on calculating the FFT for several mutually overlapping segments of the signal and averaging them. In this way a smoother and more reliable estimate of power spectral density is obtained with reduced variance of the estimate.

The calculation of the power spectral density of EEG signals can be used in various areas: sleep research and classification of sleep stages, analysis of mental states and

assessment of mental health. In mental health assessments, disorders such as depression and anxiety can be manifested through specific changes in the frequency content of the EEG signal. Analysis of power spectral density can help in the assessment and better understanding of such mental health states, and the data obtained through the analysis can serve as input features to machine learning models [15] for the classification and recognition of patterns associated with affective disorders, which will be described in more detail later in the thesis.

4.2. Spectrogram

A spectrogram is a type of visual representation of features in the time-frequency domain, in which the frequency band changes over time, and the different colors in the image reflect different values of energy in the EEG signal [16]. This representation enables improved understanding of the spatial and temporal dynamics of brain activity. Although it is in two dimensions, a spectrogram is visually interpreted as a 3D representation, since its power is shown by color as a third variable. Given that the dataset used in this thesis consists of multiple channels, such a representation can help in identifying patterns, trends and anomalies among channels, which enables a more detailed and complete analysis [15]. An example of a spectrogram is shown in Figure 4.2.1.

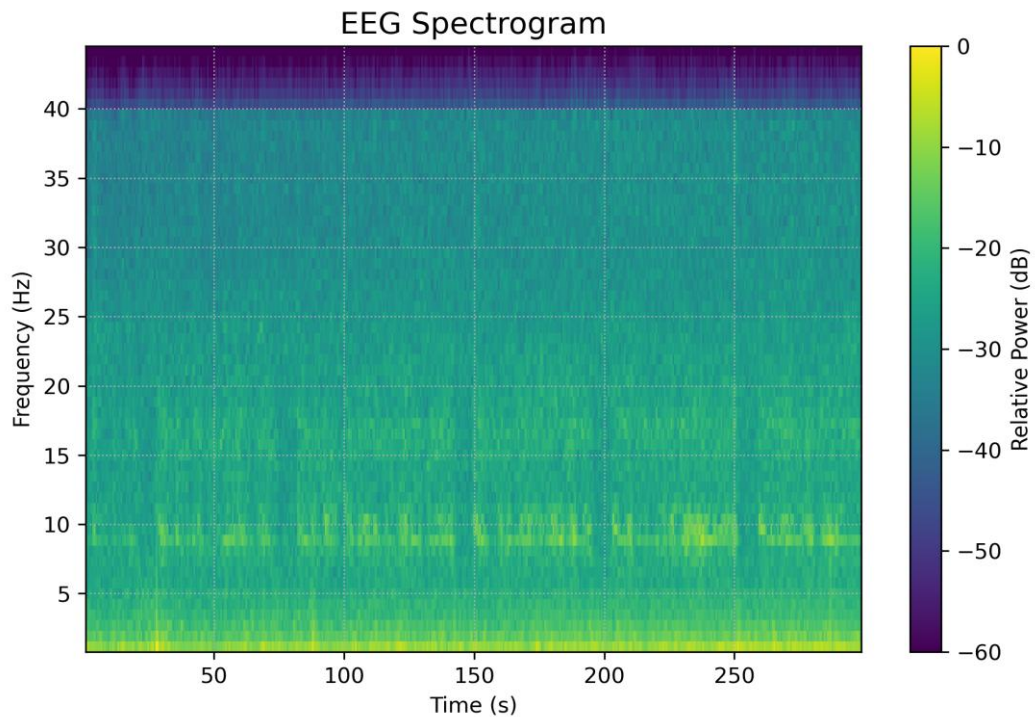


Figure 4.2.1. Spectrogram of subject 4002 displayed in Python

4.3. Topographic map

Spatial distribution is of great importance in the analysis of EEG signals because it provides valuable information about the localization and spread of brain activity. It enables researchers and doctors to map electrical activity in different areas of the brain, which improves understanding of brain processes and their mutual interactions.

By analyzing the spatial distribution of EEG signals (Figure 4.3.1.) it is possible to identify specific brain areas involved in various cognitive functions and to recognize unusual patterns of activity associated with neurological disorders. In addition, analysis of spatial distribution helps in overcoming the limitations of EEG related to relatively low spatial resolution through the application of techniques such as the *surface Laplacian transformation*, which reduces the effect of volume conduction and improves the spatial and temporal resolution of EEG [15]. Improved spatial resolution enables more precise recognition of patterns of brain activity and of differences between subjects, which can contribute to more successful identification of features associated with affective disorders.

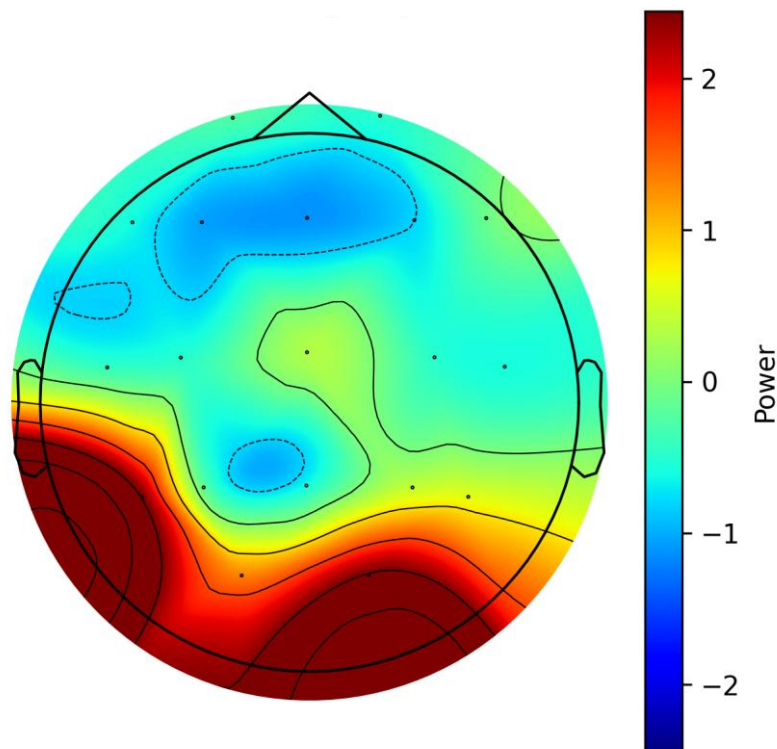


Figure 4.3.1. Topographic map of subject 4002 displayed in Python for the frequency band from 8 to 13 Hz (alpha waves)

5. Deep learning

Deep learning is a field of machine learning based on the use of multilayer artificial neural networks for learning representations from data. Unlike classical machine learning algorithms, which require features to be defined manually, deep architectures enable the automatic extraction of useful features, which enables better generalization and higher accuracy in tasks such as classification, pattern recognition or sequential analysis [17].

An artificial neural network, in the broader sense of the term, is an artificial replica of the human brain with which an attempt is made to simulate the learning process [18]. The basic building block of deep networks is the artificial neuron (Figure 5.1.), which receives several input signals, multiplies them by the corresponding weight coefficients and, through an activation function, generates an output that is passed on to the next layer [17]. The processing power of the network is stored in the strength of the connections between individual neurons, i.e. in the weights, which are arrived at by learning from the training data [18]. Furthermore, the learning of the network is based on the *backpropagation* algorithm, by which the contribution of each parameter to the total error of the model is calculated, and optimization methods such as stochastic gradient descent (*SGD*) are used to update the weights and to minimize the mentioned contribution, and thereby the total error of the model. Neural networks are very good at estimating nonlinear relationships between samples, can work with unclear or incomplete data and recognize patterns in them, and are robust to errors in the data, unlike conventional methods that assume a normal distribution of features in the input data [18].

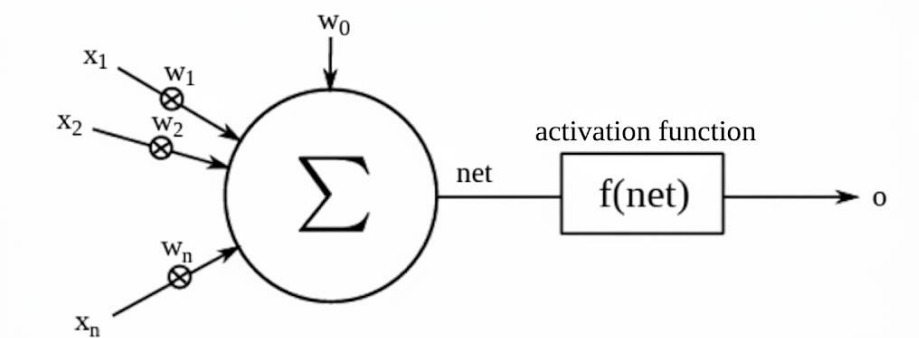


Figure 5.1. Model of an artificial neuron [18]

5.1. Convolutional neural networks

Convolutional neural networks (*CNN*) are models specialized for data with a form of structure defined by a neighborhood relation, such as a time sequence (1D), an image (2D) and a volume (3D) [17]. Generally, their name comes from the English word *convoluted*, meaning that something is very complicated. Because of this characteristic, these networks are also effective in discovering spatio-temporal patterns within EEG signals. They apply filters (kernels) to the input data with the aim of detecting local features, and in addition use subsampling layers in order to reduce dimensionality [19].

These networks, whose classical structure is shown in Figure 5.1.2., contain convolutional layers that are based on the mathematical operation of convolution, which reads:

$$h(t) = (w * x)(t) = \int_{D(w)} w(\tau)x(t - \tau)d\tau. \quad (1)$$

In the context of convolutional models, the function x is the input, the function w is the kernel, and the function h is called the feature map.

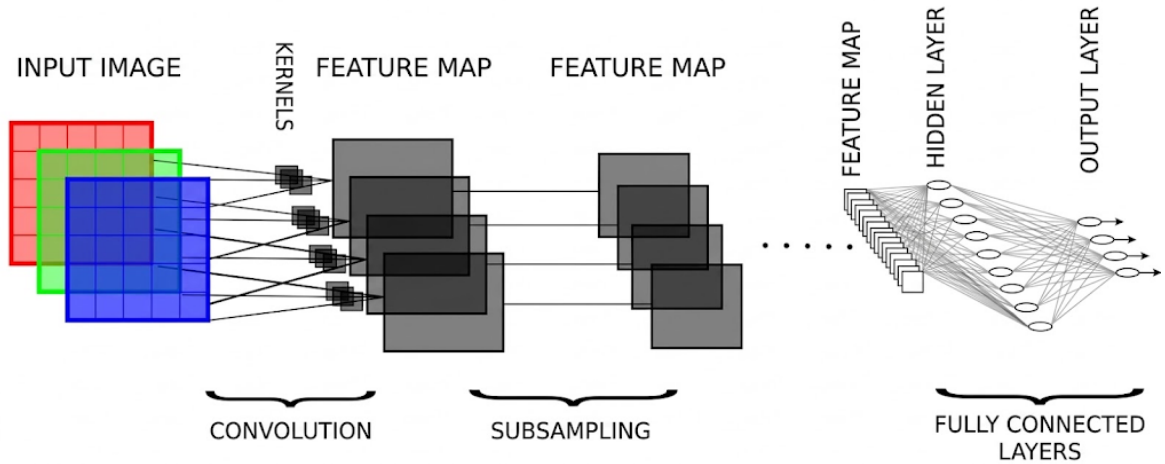


Figure 5.1.2. Classical structure of a convolutional model for image classification [17]

5.2. Grad-CAM method

When it comes to image classification, it is sometimes unclear on the basis of what information in the image the model makes its decision, so the Grad-CAM method was introduced for this purpose. Namely, the class activation mapping (CAM) method was originally

proposed as a technique of explainable artificial intelligence (XAI) that enables the identification of discriminative regions that significantly influence the predictions of convolutional neural networks used for image classification. However, CAM is limited to the last convolutional layer of the CNN, so Grad-CAM (*Gradient-weighted Class Activation Mapping*). It is an improved version of CAM that uses information contained in the gradients flowing through a selected convolutional layer in order to determine the importance of individual input elements in the decision-making process of the neural network. Thanks to this, Grad-CAM can be applied to any architecture based on convolutional neural networks.

The result of this method is a *heatmap*. To generate a heatmap specific to a particular class, Grad-CAM calculates the gradients of the output of the target class y^c with respect to the two-dimensional feature maps A^k of the selected convolutional layer, i.e. the partial derivatives $\frac{\partial y^c}{\partial A_{i,j}^k}$. These gradients are then averaged in order to obtain the weights α_k^c , which describe the importance of an individual feature map k for the target class c :

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{i,j}^k} \quad (2)$$

where Z is the number of feature maps, i.e. filters in the observed layer.

After that, the Grad-CAM heatmap is obtained by performing a weighted combination of the feature maps, followed by the application of the rectified linear unit (ReLU) activation function [20]:

$$L_{Grad-CAM}^c = \text{ReLU} \left(\sum_k \alpha_k^c \cdot A^k \right) \quad (3)$$

The ReLU activation function is applied to the linear combination of the feature maps because only those features that have a positive influence on the observed class are of interest, i.e. pixels whose intensity should be increased in order to increase the value of the output y^c . Pixels with a negative contribution probably belong to other categories present in the image. As expected, without the application of the ReLU function the localization maps sometimes also highlight areas that do not belong to the desired class, which results in poorer localization performance.

As the result of this method, a heatmap of the same dimensions as the feature maps of the corresponding convolutional layer is generated [21]. The obtained heatmap is then normalized and scaled in order to enable joint visualization with the raw EEG signal. An example of the application of Grad-CAM is shown in Figure 5.2.1.

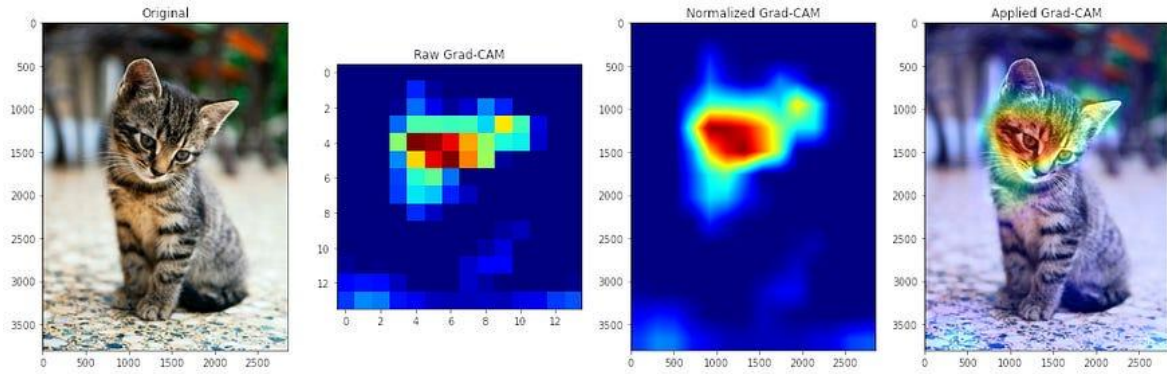


Figure 5.2.1. Application of the Grad-CAM method to an image of a cat [22]

6. Methodology

Since the dataset was already preprocessed, the next step was to generate image datasets from the EEG signals that would later be used for classification. Each representation was generated in two datasets: a smaller one and a larger one, and the extraction is described in more detail in the following subsections.

Before the generation of the image datasets itself, the resting state was extracted from each subject's recording, i.e. the first 300 s of the recording, during which the subject is in an awake and relaxed state and, according to the examiner's instructions, opens and closes their eyes. After the generation of the new, base dataset for further needs, a *Microsoft Excel* table was created in which the following are indicated: the subject's identification number, the path to the .set file of the resting state of the same subject, and the class, i.e. whether the subject was diagnosed with an affective disorder or not. The table was generated for the purpose of subsequently easier loading of the original dataset and generating the new 2D datasets. The entire process was carried out in the *Python* programming language, i.e. in a *Jupyter Notebook*.

It is important to emphasize that additional information such as accompanying displays or colour scales, axis labels, graph titles, value labels on the axes, etc., was removed from all representations. The reason for this is that the mentioned representations will later be used for training and testing deep learning models, so it is important to remove all additional elements of the graphical display that could introduce superfluous information into the system, and thereby influence the classification results themselves or contribute to overfitting of the model. The aim of this was to obtain a clean and standardized representation that emphasizes exclusively the information contained in the signal.

6.1. Extraction of 2D representations of EEG signals

6.1.1. Power spectrum

First, the smaller dataset was generated. After loading the data on the basis of the mentioned table, the power spectral density (PSD) is calculated in the frequency range from 0.5 to 45 Hz using Welch's method, which is the default method in the MNE library in

Python. Then, on the basis of the result, the PSD curve is displayed using built-in visualization functions in the same library. The obtained visualization, like the one in Figure 6.1.1.1., is then saved as a PNG file.

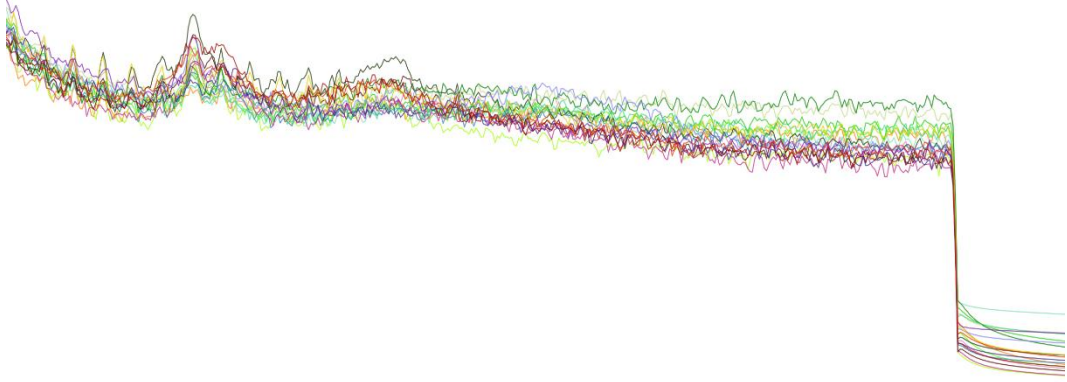


Figure 6.1.1.1. Example of a generated 2D representation of the power spectrum for subject 4002

At the end, a structured CSV file is generated that for each example, i.e. subject, contains information about their identification number, the path to the generated image and the class. In this way one image per subject was obtained, so the total number of examples in the newly generated dataset is equal to 140.

Next is the larger dataset. For the purposes of this dataset, the originally extracted resting state was divided into five equal periods and 2D representations were displayed for each of the individual periods. In this way an expanded dataset was obtained, in which, instead of 140 examples, 700 are considered. The aim of the mentioned method was the augmentation of the dataset, since for the purposes of classification using deep learning the largest possible datasets are needed so that the model can learn as much as possible. Each EEG signal, after it is loaded, is first divided in the following way:

$$segment\ duration = \frac{overall\ duration\ of\ the\ resting\ state}{number\ of\ segments}. \quad (4)$$

Each segment is extracted using the function `crop (tmin, tmax)`. After extraction, the power spectral density in the frequency range from 0.5 to 45 Hz is then calculated for each segment, and the PSD curve is displayed again. All additional labels, i.e. information, are removed from the representation and they are saved as PNG files, with the generation of a CSV table that contains the subject's identification number, the segment number, the path to

the obtained representation and the class. Figures 6.1.1.2. – 6.1.1.6. show the obtained power spectra of all five segments for one subject.

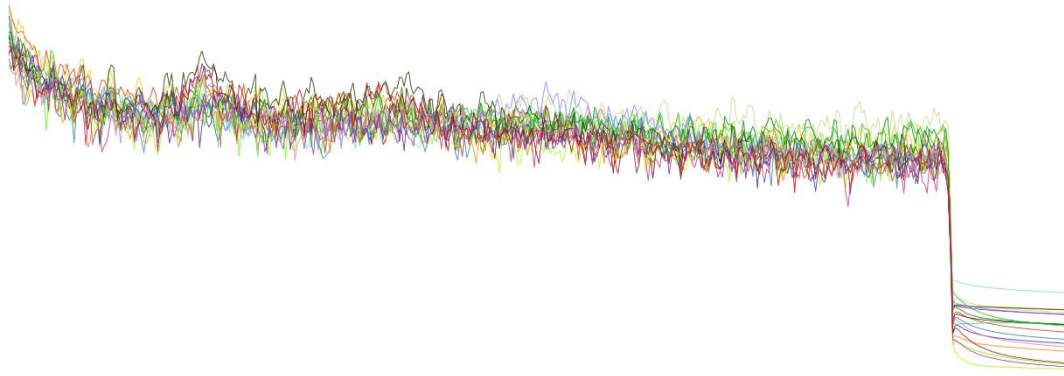


Figure 6.1.1.2. Power spectrum for segment 0 of subject 4002

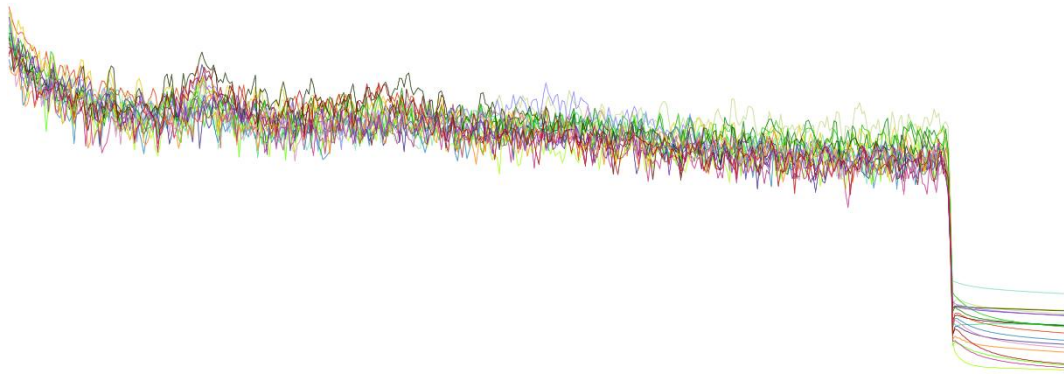


Figure 6.1.1.3. Power spectrum for segment 1 of subject 4002

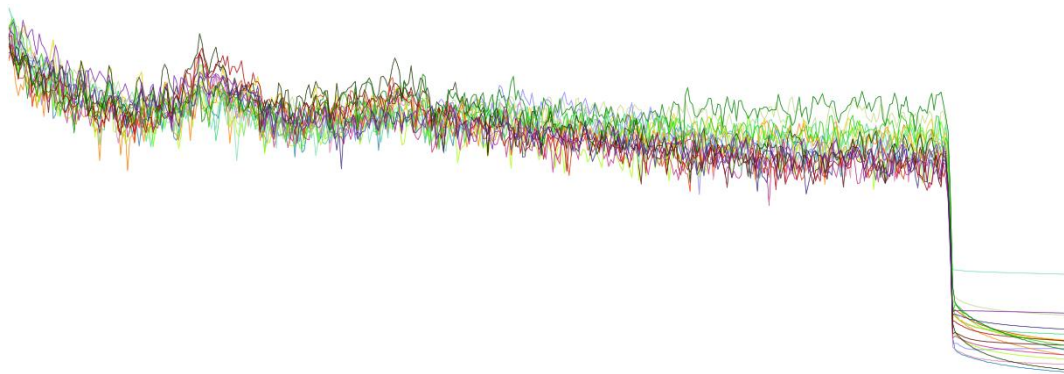


Figure 6.1.1.4. Power spectrum for segment 2 of subject 4002

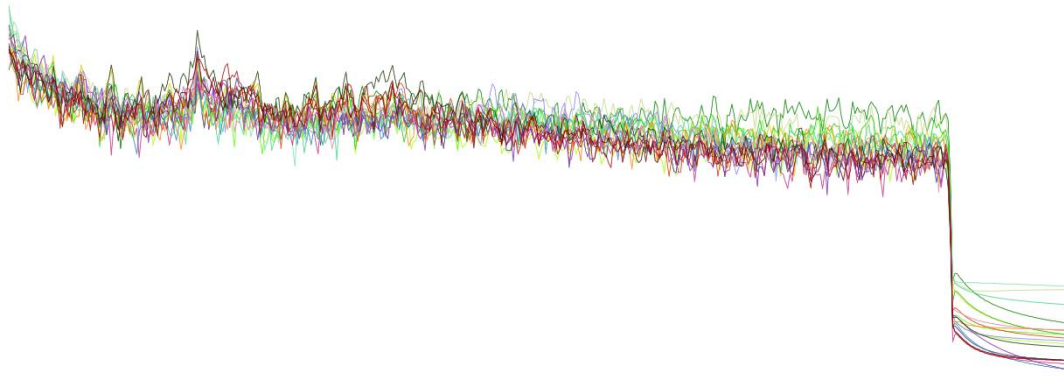


Figure 6.1.1.5. Power spectrum for segment 3 of subject 4002

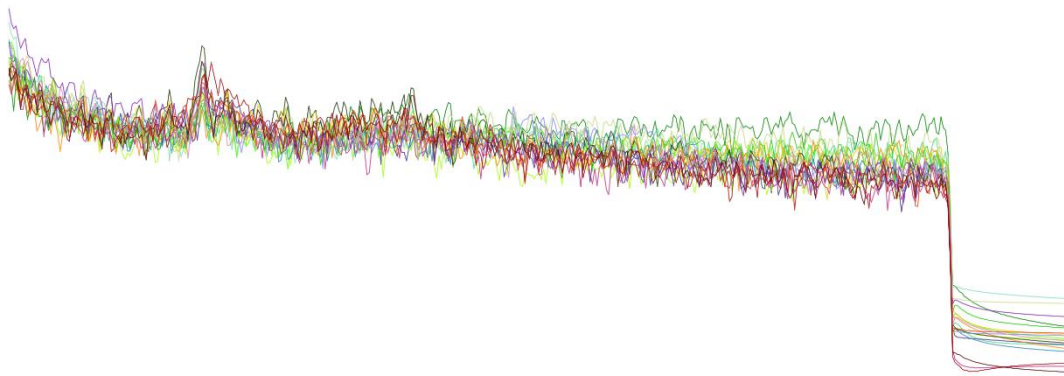


Figure 6.1.1.6. Power spectrum for segment 4 of subject 4002

6.1.2. Spectrogram

After loading the data, the 2D spectrograms of the smaller dataset were extracted first. The sampling frequency, which is needed for further time-frequency analysis, is retrieved from the initial data. After applying a band-pass filter in the range from 0.5 to 45 Hz, Z-normalization is carried out for each channel separately in order to standardize the amplitudes among channels and reduce inter-channel variability. Next, for each EEG channel the *short-time Fourier transform* (STFT) is computed using the function `spectrogram()` from the SciPy library. The parameters used were `nperseg = 256` (number of samples per window), `noverlap = 128` (50% overlap between adjacent windows for smoother transitions), `mode = 'psd'` (the output represents power spectral density) and `scaling = 'density'` (normalized power per unit of frequency). The obtained 2D representation is a time-frequency representation that shows the changes in power of individual frequency components over the duration of the recording. The obtained spectrograms are then averaged over all EEG signals, which

reduces the influence of noise and emphasizes global patterns of brain activity. Then the power values are converted to a logarithmic scale (5):

$$P_{dB} = 10 \log_{10}(P + 10^{-10}), \quad (5)$$

which enables better visibility of weaker frequency components and emphasizes patterns that would otherwise be difficult to notice. This is followed by normalization (6) to the maximum value in order to obtain a consistent intensity scale for the entire dataset:

$$P_{norm} = P_{dB} - \max(P_{dB}). \quad (6)$$

The spectrogram is then displayed as a two-dimensional *heatmap* in which the x-axis denotes time (s), the y-axis denotes frequency (from 0.5 to 45 Hz), and the colour denotes the signal power expressed in decibels. The 'viridis' display was used, and the display range was fixed at from -60 to 0 dB, which is the usual standard in EEG analysis. All additional labels are removed and each spectrogram is saved as an image (example in Figure 6.1.2.1.). A CSV file is also generated with information about the subject's identification number, the path to the obtained spectrogram and the class.

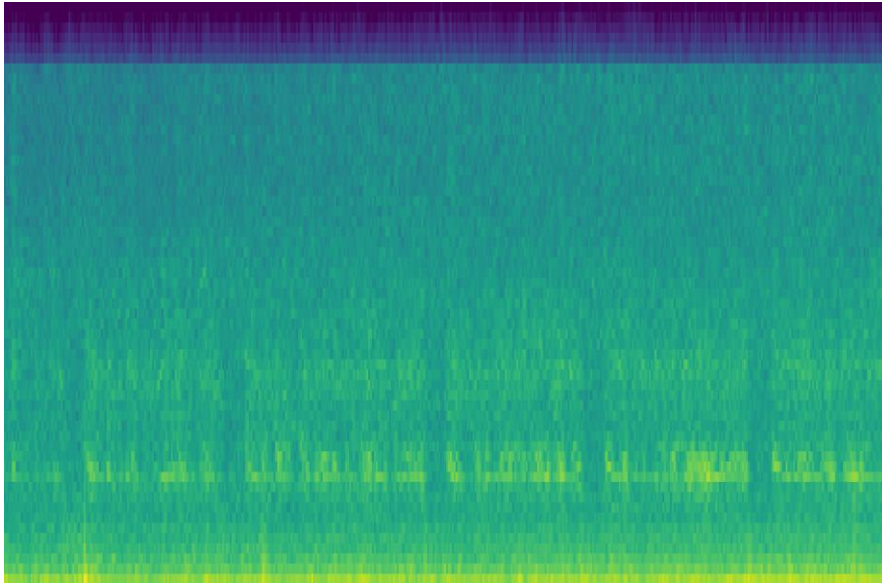


Figure 6.1.2.1. Spectrogram generated for subject 4002

Next, the augmented dataset of spectrograms was generated, in which each subject's EEG signal was again divided into five equal parts as described in expression (4). After division into segments, the further procedure is the same as that for the smaller spectrogram dataset, already described. The only difference is that in the end a dataset of 700 representations is obtained, as opposed to the 140 in the smaller dataset, and the CSV table consists of

information about the subject's identifier, the segment on the basis of which the spectrogram was generated, the path to it and the class. Figures 6.1.2.2. – 6.1.2.6. show the spectrograms obtained by segmenting the EEG signal of one subject.

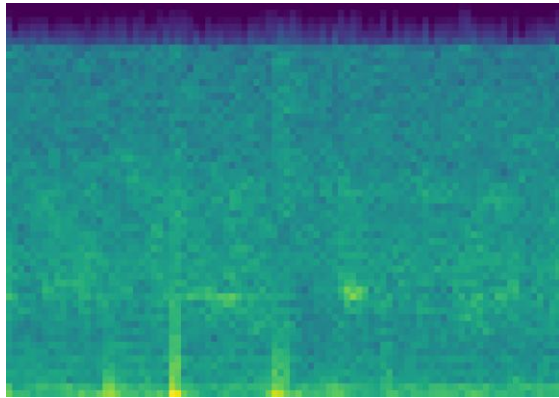


Figure 6.1.2.2. Spectrogram of segment 0 of subject 4009

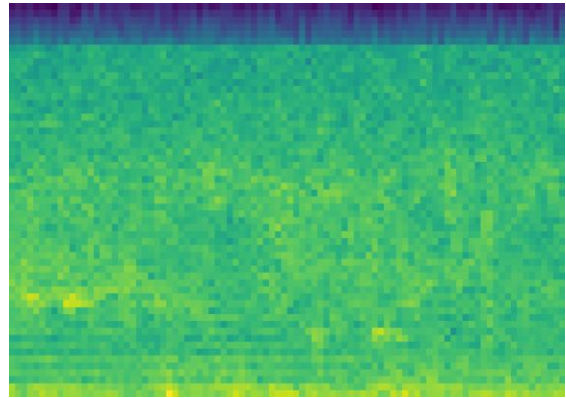


Figure 6.1.2.3. Spectrogram of segment 1 of subject 4009

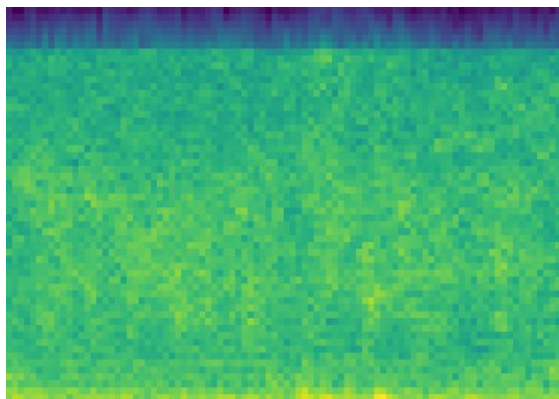


Figure 6.1.2.4. Spectrogram of segment 2 of subject 4009

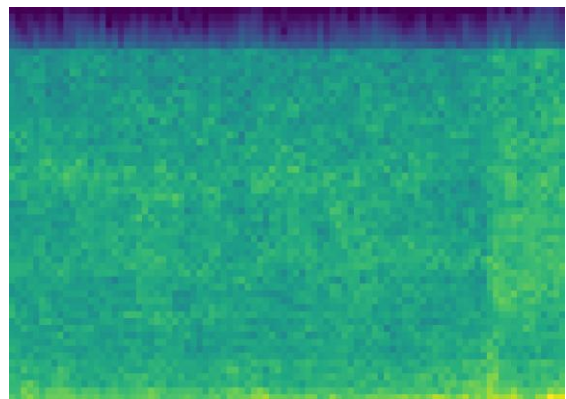


Figure 6.1.2.5. Spectrogram of segment 3 of subject 4009

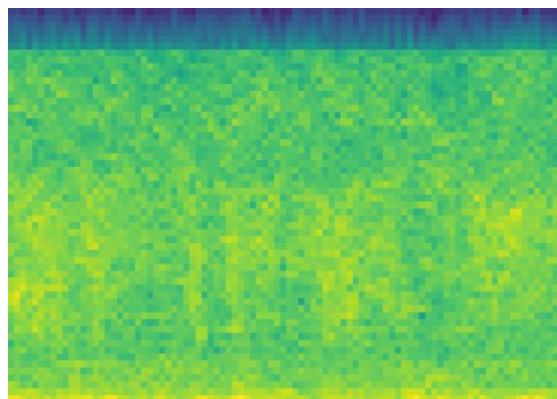


Figure 6.1.2.6. Spectrogram of segment 4 of subject 4009

6.1.3. Topographic maps

When it comes to topographic maps, the smaller dataset was generated first. After loading the signals, it was necessary to rename the electrodes so that they would be compatible with the standard EEG electrode nomenclature. Therefore, for all electrodes that had '-Ref' in their name, that part was removed. The standard 10-20 electrode arrangement was loaded, which is important so that all representations have a universal electrode arrangement. Filtering in the range from 0.5 to 45 Hz followed, and the frequency ranges of the brain waves were defined: delta (0.5 – 4 Hz), theta (4 – 8 Hz), alpha (8 – 13 Hz), beta (13 – 30 Hz) and gamma (30 – 45 Hz). Calculation of power spectral density using Welch's method followed, with the parameters: $n_fft = 1024$ (frequency resolution), $n_overlap = 512$ (50% window overlap), $n_per_seg = 1024$ (window length) and $average = 'median'$ (median instead of the arithmetic mean for greater robustness to artifacts and extremes). From the obtained PSD, the power values within each frequency band are extracted for each EEG signal, and for each frequency band multi-stage normalization is then carried out in order to improve comparability between subjects. First, the average PSD value within the corresponding frequency range was determined for each channel. Then the power of each band was expressed as a share of the total EEG power (7)

$$P_{rel} = \frac{P_{band}}{P_{total}}, \quad (7)$$

which removes the influence of the total amplitude of the signal and redirects attention to smaller values and oscillations in power. Z-normalization was then applied, which achieves a consistent scale between subjects. This was followed by limiting outlying values by means of the clip function to ± 3 standard deviations. This reduced the influence of isolated channels, while most of the information from the normal range of values is retained. For each frequency band a topographic map is generated using the function *plot_topomap()* from the MNE library. In the display, the x-axis and y-axis represent the physical positions of the electrodes on the scalp, and the colour intensity represents the normalized power of the brain waves. The colour map *jet* was used, in which red denotes higher power and blue denotes lower power, and six contour lines are displayed for easier interpretation of the transitions between colours. All additional information was again removed from the images, so the final result is shown in Figures 6.1.3.1. – 6.1.3.5. The topographic maps were saved in the form of PNG files and a CSV table was generated with the following data: the subject's

identification number, the name of the frequency band, the path to the obtained representation and the class. The obtained dataset has 700 examples, since for each subject topographic maps were extracted for the previously mentioned frequency intervals that represent particular brain waves.

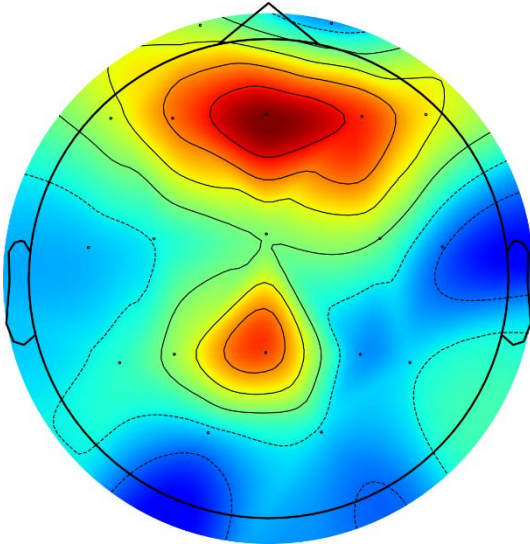


Figure 6.1.3.1. Topographic map of subject 4002 for delta waves

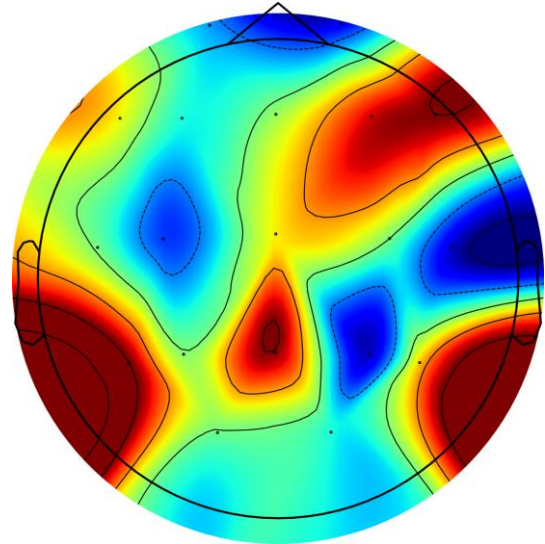


Figure 6.1.3.2. Topographic map of subject 4002 for theta waves

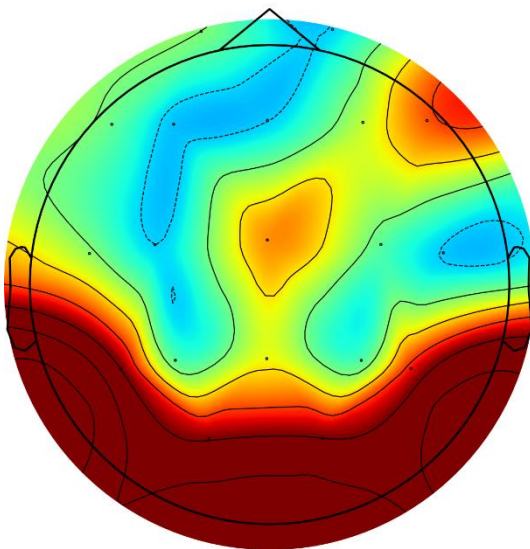


Figure 6.1.3.3. Topographic map of subject 4002 for alpha waves

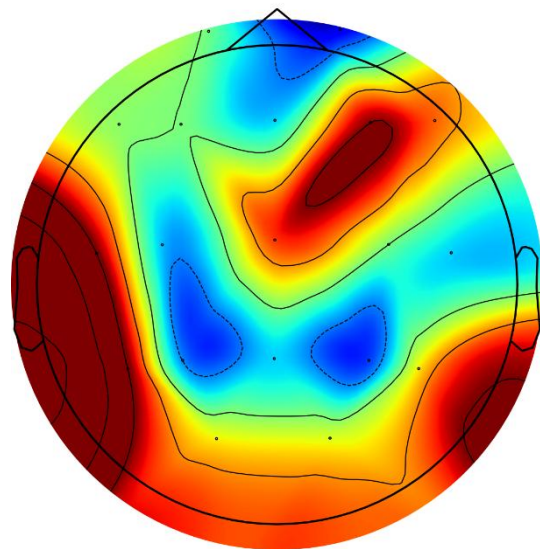


Figure 6.1.3.4. Topographic map of subject 4002 for beta waves

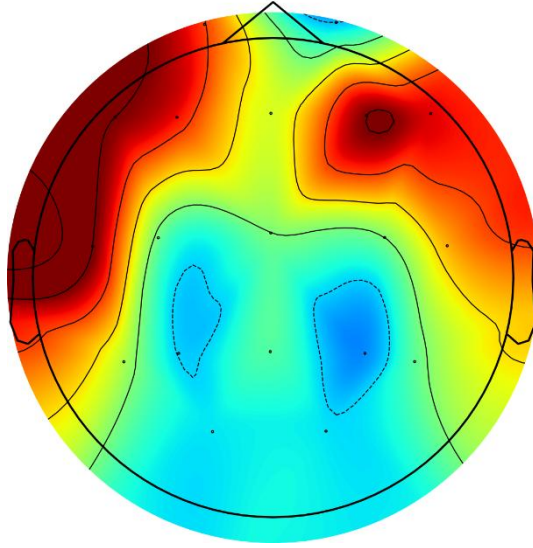


Figure 6.1.3.5. Topographic map of subject 4002 for gamma waves

Next, the larger dataset was generated, in which one subject's EEG signal was again segmented in the way shown in expression (4). After that, for each of the five time segments the same processing described for the smaller dataset is carried out: calculation of the PSD, extraction of the power of each of the frequency ranges, normalization of relative power, Z-normalization and limiting of extreme values, and generation of topographic maps. In this way one topographic map is generated for each segment for each frequency band. By this principle a dataset of a total of 3500 images was obtained.

6.2. Classification of 2D representations of EEG signals

The procedure for classifying the 2D representations is generalized, so the differences for individual datasets will be pointed out in the following subsections, while here the procedure used for all datasets will be described. After the data are loaded, the preprocessing procedure follows, and once the image representations have been preprocessed, their conversion into a tensor form suitable for processing with the *PyTorch* tool follows. As the final step of preprocessing, normalization of pixel intensity was carried out using a mean value of 0.5 and a standard deviation of 0.5 so that pixel values are mapped to approximately the range from -1 to 1, which achieves numerical stability during training and speeds up the convergence of the model. The dataset was then divided into three disjoint sets: a training set, a validation set and a test set. 60% of the data was set aside for training, 20% for the validation set and 20% for the test dataset. For the purposes of training, the data are

distributed into *mini-batches* and *shuffled*, while the validation and test sets are used without permutation.

For classification, a simple convolutional neural network was used, consisting of three consecutive convolutional layers with a gradual increase in the number of features (16, 32 and 64 filters). Each convolutional layer uses 3x3 kernels with *padding* so that the spatial dimensions of the features are preserved. After each convolutional layer the *ReLU* activation function is applied, as well as a *max-pooling* operation that reduces spatial dimensionality and increases the model's robustness to local translations. After the convolutional part, adaptive global pooling of features (*adaptive average pooling*) to a fixed size of 4x4 is used, which ensures the model's independence of the input dimensions after scaling. The extracted features are then flattened and passed to the *fully connected* part of the network, which consists of a hidden layer with 128 neurons and an output layer with two classes. *Dropout* regularization with a probability of 0.5 is applied to the hidden layer in order to reduce overfitting of the model.

The model is trained as a binary classifier using cross-entropy (*CrossEntropyLoss*). The Adam optimizer with a given *learning rate* is used to optimize the parameters. This combination of loss and optimizer enables stable and efficient learning on image representations of EEG data. Training was carried out over 30 epochs, and in each epoch the model passes through all the *mini-batches* of data, during which prediction is performed, the loss function is calculated and the network parameters are updated using the optimization algorithm. During training, the performance of the model is continuously monitored on the validation set in order to assess the ability to generalize and to prevent overfitting. The accuracy of the model on the validation set is calculated after each epoch as the ratio of correctly classified samples to the total number of samples. After training is finished, the model is switched to evaluation mode and the classification accuracy of the model is assessed.

6.2.1. Power spectrum

During the preprocessing of the smaller dataset, the images are converted to *gray-scale*, which reduces the number of input features and the computational complexity of the model. Then all images are scaled to a resolution of 128x256 pixels in order to ensure consistent dimensionality of the input to the convolutional neural network. The data were divided into training, validation and test sets using the function *train_test_split()*, where the

split was done at the level of images. The split was carried out in a stratified manner according to class labels in order to preserve the same class distribution in all subsets. Such a split ensures a balanced ratio of classes in all phases of model evaluation, which reduces the risk of bias in the assessment of the classes. The learning rate was 0.001, and the training set was distributed into 16 mini-batches of training data.

The preprocessing procedure for the larger power spectrum dataset also included conversion to grayscale and scaling of all images to a resolution of 128x256 pixels. The set was divided into training, validation and test sets using the function *GroupShuffleSplit()*, where the split was done at the level of subjects. This was done in order to prevent images of the same subject from appearing in different datasets, which prevents information leakage (*data leakage*) between the training and evaluation phases of the model and enables a more realistic assessment of the model's ability to generalize. The learning rate was 0.001 and the training set was divided into 16 mini-batches of training data.

6.2.2. Spectrogram

The preprocessing procedure for the smaller spectrogram dataset was as follows: the images were converted to grayscale and scaled to a resolution of 256x256 pixels in order to ensure universal input dimensionality for all data. The data were divided into training, validation and test sets using the function *train_test_split()*, where the split was done at the level of images. The split was carried out in a stratified manner according to class labels in order to preserve the same class distribution in all subsets. Such a split ensures a balanced ratio of classes in all phases of model evaluation, which reduces the risk of bias in the assessment of the classes. The learning rate used in this classification was 0.002, and the training set was divided into 16 mini-batches of training data.

In the preprocessing of the larger dataset, the images were likewise converted to grayscale, scaled to 256x256 pixels and divided into subsets using the function *GroupShuffleSplit()*, where the split was again done at the level of subjects, so that images of the same subject would not end up in different sets. The learning rate was 0.002 and the training set was divided into 16 mini-batches of training data.

6.2.3. Topographic maps

In the classification of topographic maps, the main difference is that one sample, i.e. one input to the CNN, represents a set of five topographic maps of the same subject. These topographic maps correspond to frequency bands for the same subject or the same segment of a subject, depending on the dataset. The maps are predefined in a universal order so that the model always receives the same structure of input data, and that order was defined as follows: delta, theta, alpha, beta, gamma waves.

In the preprocessing of the smaller dataset, the images were loaded in the form of a colour scale (RGB), scaled to a resolution of 256x256 pixels and then divided using the function *StratifiedGroupKFold()* in order to preserve the class ratio within each subset. Given that the model receives as input five topographic maps of the same subject, each map passes through a shared convolutional feature extractor (*shared feature extractor*), after which the extracted features of all frequency bands are merged into a single feature vector that is used for the final classification, which yielded a dataset of 140 examples. The learning rate was 0.001 and the training subset was divided into 16 mini-batches of training data.

For the larger dataset the representations were also left in the original colour scale (RGB) and were transformed into images with a pixel ratio of 265x265. For this dataset, the representations of all frequency ranges of the same segment of the same subject were grouped together. The same principle as for the smaller dataset was applied, but here, after grouping, a final dataset of 700 examples was obtained, which was then divided into training, validation and test sets. The split of the dataset was carried out at the level of subjects, whereby all segments of the same subject are located exclusively in one set, which prevents information leakage between sets. The split was done using the function *StratifiedGroupKFold()*, and the training subset was additionally divided into mini-batches of 8 examples because of the increased memory demands of the multidimensional input of this dataset. The model also uses a shared convolutional feature extractor for all five frequency bands within each time segment, and the extracted features are concatenated and used further for classification. The learning rate was 0.001.

6.3. Implementation of the Grad-CAM method

The Grad-CAM method was applied for the interpretation of the decisions of the convolutional neural network, and it was used exclusively on models trained on the topographic map datasets of EEG signals (the smaller and the larger dataset). The input to the model consisted of a set of five topographic maps corresponding to the frequency bands, whereby all maps of the same subject/segment were evaluated jointly in a single prediction.

In the implementation for the purposes of this thesis, the Grad-CAM analysis was carried out on the last convolutional layer of the CNN, immediately before the fully connected part of the network, which means that the method was based on the global prediction of the model and not on individual bands. In general, the result of this procedure is a heatmap, and here a separate Grad-CAM map was generated for each frequency band, which provided insight into the contribution of individual frequency components of the EEG signal to the final classification. Each obtained map was then normalized to the range $[0, 1]$ in order to enable comparability between frequency bands. Finally, the visualization is shown as the original topographic maps side by side with the corresponding Grad-CAM activation maps for each frequency band, where the original topographic maps were first converted into black-and-white images, and the activation maps are shown in colour over them, so that the areas important for the model's decision-making can be more easily noticed.

7. Results

7.1. Classification results

Accuracy and the F1 measure were used as measures of the model's classification accuracy, along with a graphical display called the confusion matrix. The confusion matrix consists of four cells: TP, or *true positive* classification, FP, or *false positive* classification, FN, or *false negative* classification, and TN, or *true negative* classification. The TP cell contains examples that were correctly classified as positive examples, the FP cell contains examples that were incorrectly classified as positive examples, the FN cell contains examples incorrectly characterized as negative, and analogously the TN cell contains examples correctly classified as negative.

Accuracy is a measure calculated according to expression (8):

$$Acc = \frac{TP + TN}{TP + TN + FP + FN}. \quad (8)$$

The F1 measure takes into account two different measures, precision and recall, so it is necessary to mention them as well. Precision (9) is defined as follows:

$$P = \frac{TP}{TP + FP}, \quad (9)$$

and recall (10) reads:

$$R = \frac{TP}{TP + FN}. \quad (10)$$

Precision and recall give us different information, so in practice the F1 measure (11) is often used, defined as the harmonic mean of precision and recall:

$$F1 = \frac{2}{\frac{1}{P} + \frac{1}{R}} = \frac{2 \cdot PR}{P + R}. \quad (11)$$

7.1.1. Power spectrum

For the smaller dataset, the confusion matrix shown in Figure 7.1.1.1. was obtained.

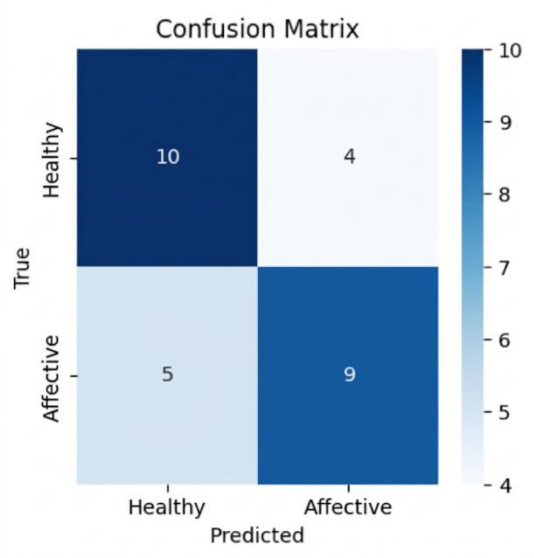


Figure 7.1.1.1. Confusion matrix of the classification of the power spectrum, smaller dataset

The accuracy for this dataset is 67.86%, and the F1 measure is 66.67%, which indicates a moderate ability of the model to distinguish the classes on the basis of the global representation of the power spectrum of the EEG signal.

On the larger, segmented dataset an accuracy of 66.43% was achieved, while the F1 measure was 71.17%. Although the accuracy is slightly lower compared with the smaller set, the increased F1 measure indicates a better balance of the model in recognizing the classes, especially in the context of a more complex data structure. The confusion matrix is shown in Figure 7.1.1.2.

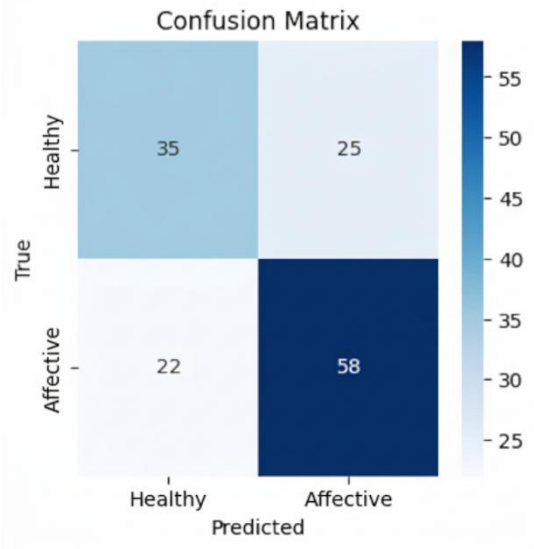


Figure 7.1.1.2. Confusion matrix of the classification of the power spectrum, larger dataset

7.1.2. Spectrogram

With spectrograms, an improvement in performance on the larger dataset was recorded. On the smaller dataset an accuracy of 67.86% was achieved with an F1 measure of 64.00%, while on the larger dataset the accuracy rose to 71.43% and the F1 measure to 72.60%. The confusion matrix of the smaller dataset is shown in Figure 7.1.2.1, and that of the larger dataset in Figure 7.1.2.2.

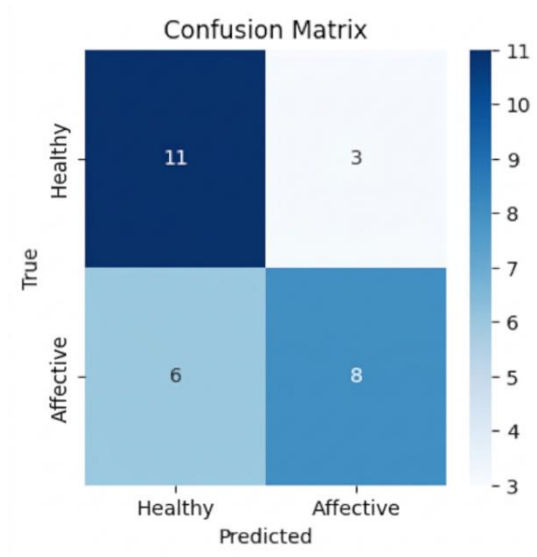


Figure 7.1.2.1. Confusion matrix of the classification of spectrograms, smaller dataset

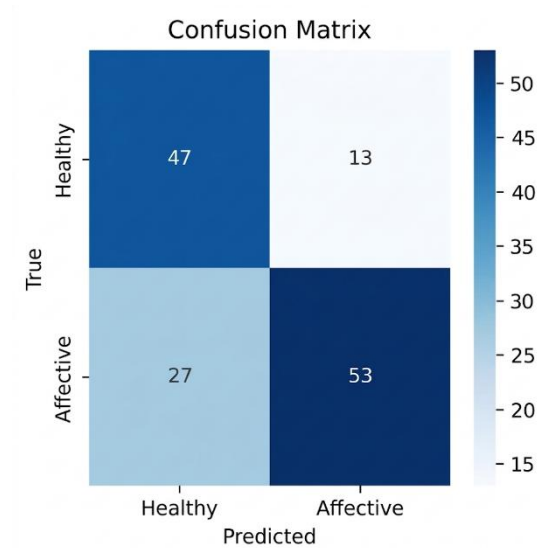


Figure 7.1.2.2. Confusion matrix of the classification of spectrograms, larger dataset

The increase in the F1 measure of the larger dataset relative to the smaller set points to a better balance of classification between the classes, i.e. a reduction of bias towards the dominant class. At the same time, the rise in accuracy suggests that the model generalizes better when trained on a larger number of samples, which is expected given the greater variability of the data in the representations themselves, which include both temporal and frequency information.

7.1.3. Topographic maps

With topographic maps, significantly better performance was achieved compared with power spectra and spectrograms.

Classification of the smaller dataset of topographic maps gave an accuracy equal to 85.71%, an F1 measure of 88.89% and the following confusion matrix of the images (Figure 7.1.3.1):

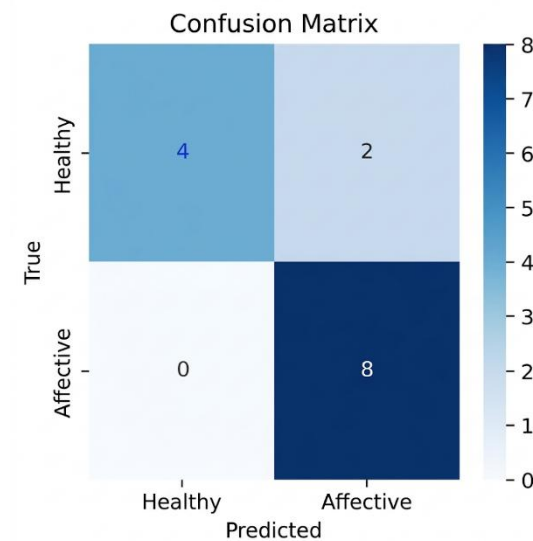


Figure 7.1.3.1. Confusion matrix of the classification of topographic maps, smaller dataset

In the classification of the larger dataset the following results were obtained: an accuracy of 88.57%, an F1 of 89.47% and the confusion matrix in Figure 7.1.3.2.

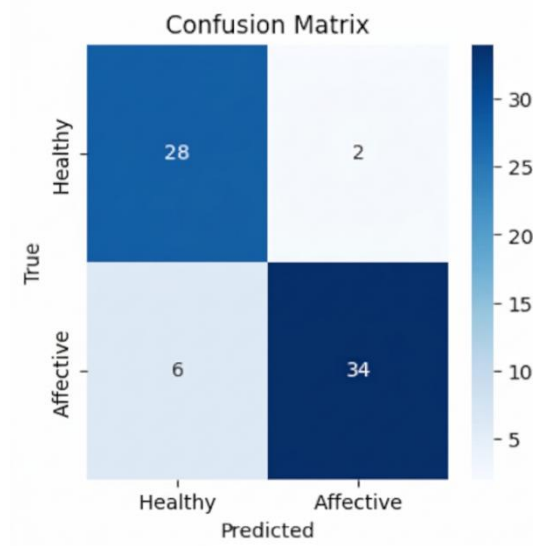


Figure 7.1.3.2. Confusion matrix of the classification of topographic maps, larger dataset

These results show a stable and consistent increase in performance with the enlargement of the dataset, with a particularly pronounced high F1 result, which indicates a very good balance of classification between the classes. Such an outcome gives indications that the spatial representation of the EEG signal in the form of topographic maps contains more

informative and more discriminative features for the classification task compared with frequency and time-frequency representations.

Finally, an overview of all results is shown in Table 1.

Table 1. Overview of all classification results

Dataset	Number of generated images	Number of examples for classification	Accuracy	F1
Power spectrum – smaller dataset	140	140	67.86%	66.67%
Power spectrum – larger dataset	700	700	66.43%	71.17%
Spectrogram – smaller dataset	140	140	67.86%	64.00%
Spectrogram – larger dataset	700	700	71.43%	72.60%
Topographic maps – smaller dataset	700	140	85.71%	88.89%
Topographic maps – larger dataset	3500	700	88.57%	89.47%

7.2. Results obtained with the Grad-CAM method

By applying Grad-CAM to the smaller dataset of topographic maps, the results shown in Figure 7.2.1. and Figure 7.2.2. were obtained. The individual in Figure 7.2.1. was diagnosed with an affective disorder and the model also placed them in that class in classification, while the individual in Figure 7.2.2. was not diagnosed with an affective disorder and was placed in the wrong class during classification.

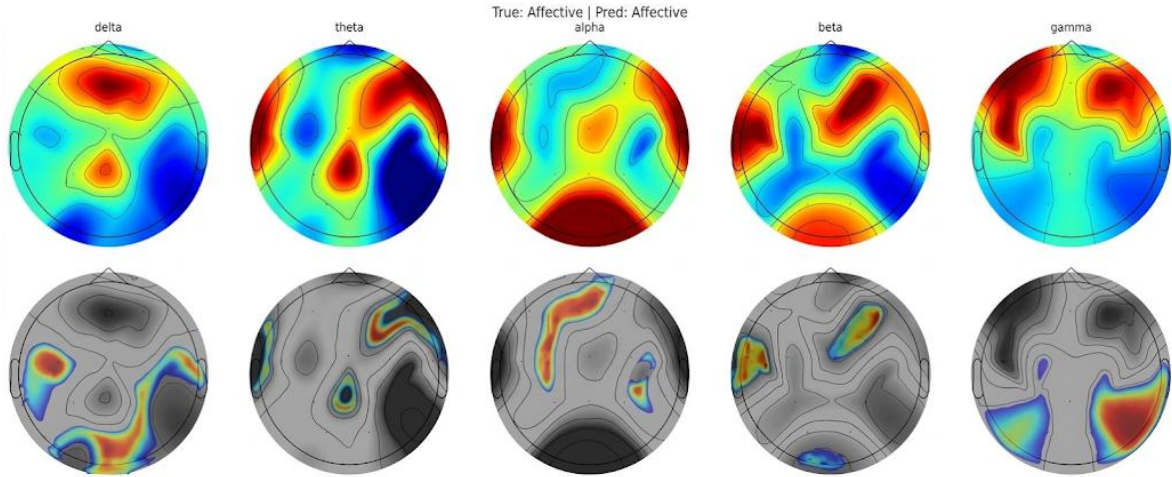


Figure 7.2.1. Side-by-side original topographic maps and results of the Grad-CAM method for subject 4002

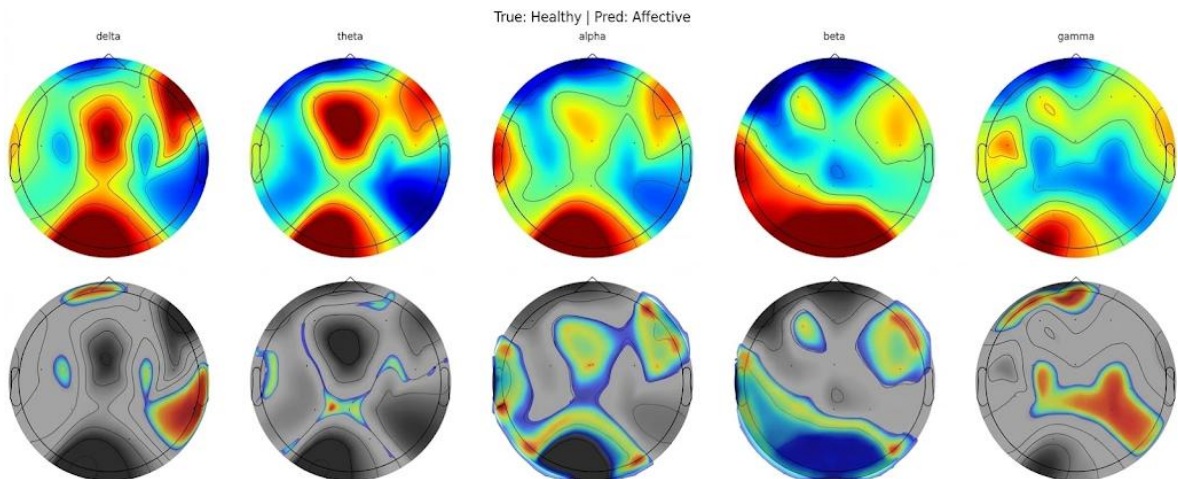


Figure 7.2.2. Side-by-side original topographic maps and results of the Grad-CAM method for subject 6745

By applying the Grad-CAM method to the larger dataset of topographic maps, the results shown in Figures 7.2.3. – 7.2.7. were obtained for one subject.

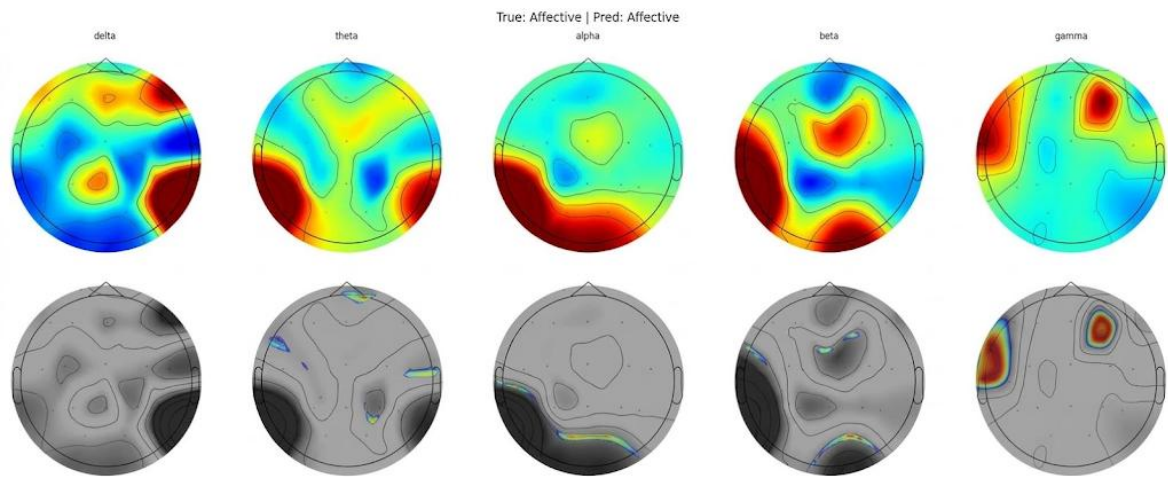


Figure 7.2.3. Side-by-side original topographic maps and results of the Grad-CAM method for segment 1 of subject 4002

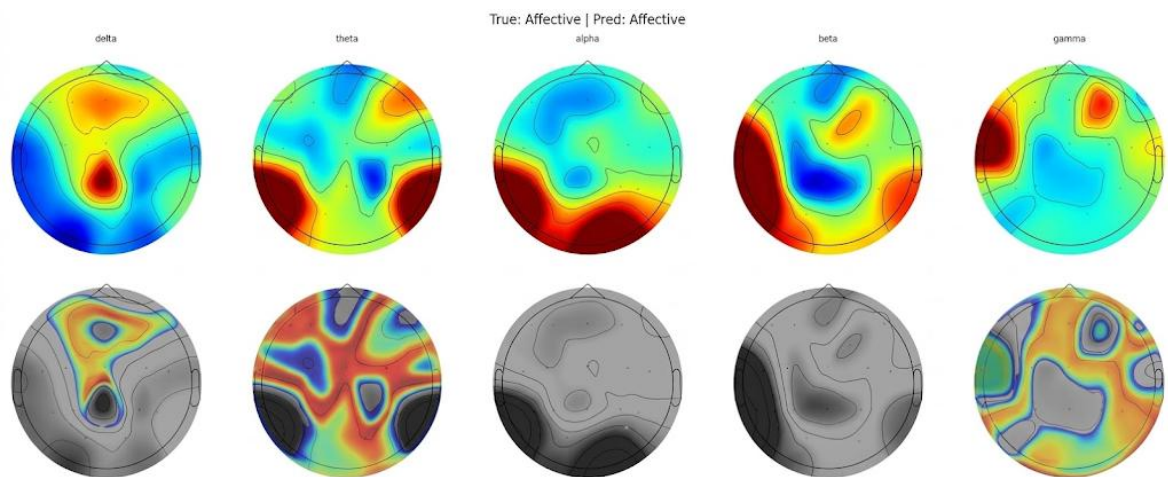


Figure 7.2.4. Side-by-side original topographic maps and results of the Grad-CAM method for segment 2 of subject 4002

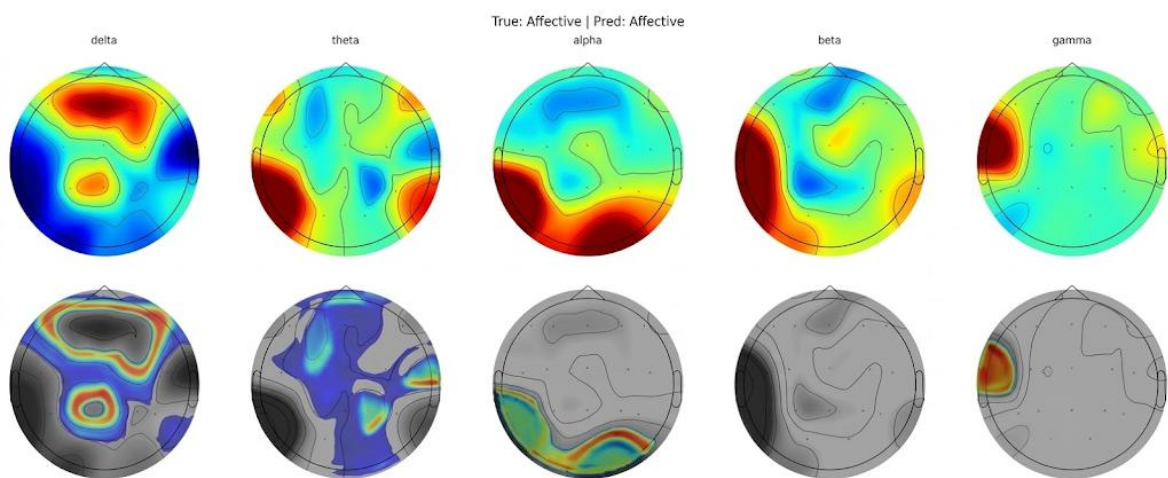


Figure 7.2.5. Side-by-side original topographic maps and results of the Grad-CAM method for segment 3 of subject 4002

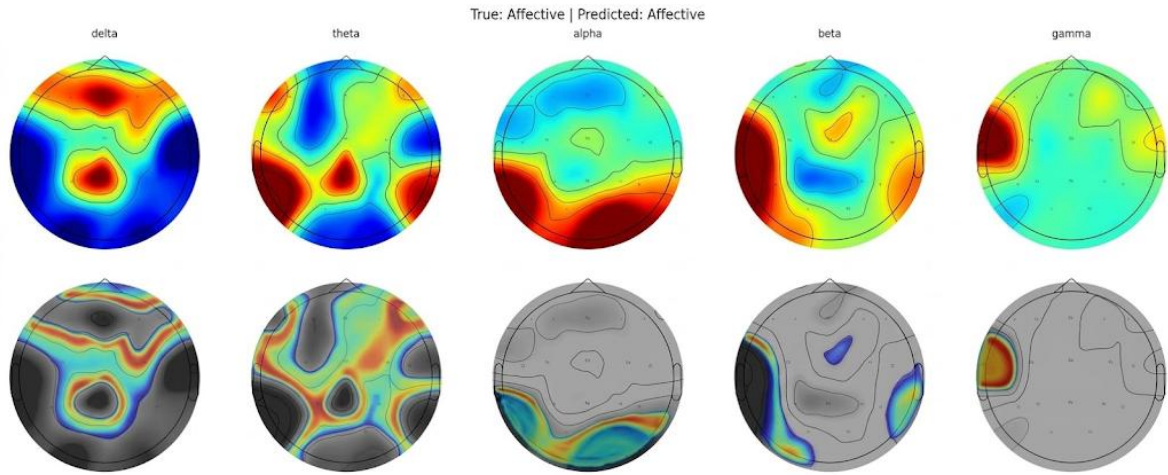


Figure 7.2.6. Side-by-side original topographic maps and results of the Grad-CAM method for segment 4 of subject 4002

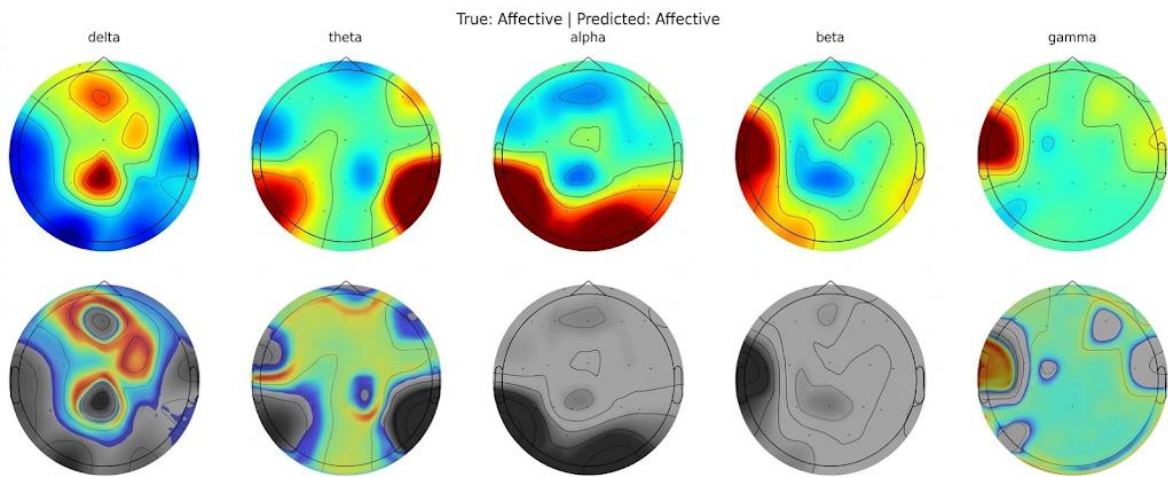


Figure 7.2.7. Side-by-side original topographic maps and results of the Grad-CAM method for segment 5 of subject 4002

On the basis of the obtained results it can be concluded that, when making a decision, the model gives the most importance to those areas of the topographic maps where the change in signal strength is most drastic. As can be seen in the displays, the areas that are turquoise, yellow or orange on the original topographic maps are marked most strongly as the areas on the basis of which the model made its decision. It is interesting that the model does not make decisions on the basis of the areas where the signal is strongest, as might perhaps initially have been expected.

Conclusion

Affective disorders are a type of mental disorder that records a constant increase in the number of people affected, and it is necessary to find a procedure by which they can be diagnosed in a faster and more objective way. One idea of how this can be achieved is the application of artificial intelligence to EEG signals. The currently most widespread methods of classifying brain signals are based on extracting features of the signal and applying machine or deep learning models to them. However, the question arises whether there is a possibility of a different interpretation of signals. Namely, the field of deep learning based on signals is much less explored compared with the field based on images. From this realization the idea for this thesis arose, in which various image-based 2D representations of EEG signals and the accuracy of their classification were investigated, as well as how the model makes its classification decision, by means of the Grad-CAM method.

Within this thesis the focus was on generating images of power spectra, spectrograms and topographic maps of EEG signals, which were then passed through a simple convolutional network in order to find out how well the image representation alone can be classified. In total, six datasets were generated, one smaller and one larger, segmented dataset for each type of representation.

The results obtained for the power spectrum representations are the worst, but this is expected, since it is an ordinary line graph that contains exclusively frequency information, without the temporal component of the signal. The results were slightly improved by expanding the dataset, but not enough for this representation to be seriously considered for further research. Spectrograms showed an improvement in results, especially with the larger dataset, which indicates that including time-frequency dynamics contributes to better discriminative power of the model. The best results are achieved by topographic maps, which suggests that the spatial distribution of the brain-activity signal carries the most relevant information when it comes to the classification task in this thesis.

By repeatedly running the classification process, variations in the achieved results were observed, especially with the smaller datasets. In this way the direct influence of dataset size on the stability of model learning is visible. Since the smaller dataset was additionally divided into subsets for training, validation and testing, the number of examples in individual

subsets became relatively small. Under such conditions even a different classification of one or several examples can significantly affect the overall accuracy and F1 measure of the model. In addition, the results depend on the data split itself, on the random initialization of the network weights, on the order of mini-batches during training, and on the applied regularization techniques such as *dropout*.

Despite the mentioned variations, the results consistently show better performance on larger datasets for all the observed types of EEG signal representation. This confirms that a larger quantity and diversity of data contributes to better generalization of the deep learning model. Segmentation of the signal and the use of multiple representations per subject also contributed to more stable learning of the model, although their effect depended on the type of representation used.

The implementation of the Grad-CAM method gave insight into on the basis of which aspect of the topographic map the model makes its decision, and on the basis of the obtained results it is evident that the model focuses on the areas of the image with the most pronounced changes in signal power.

The obtained results give an interesting picture of the possible further research into the classification of 2D representations of EEG signals, especially topographic maps, whereby an accuracy of 88.57 % and an F1 of 89.47 % on the larger dataset are very good classification results, especially given the nature of the problem in question. Further research could include expanding the original dataset by recording new recordings in order to ensure enough data for more robust model learning. In addition to expanding the dataset, stabilization of the classification results can also be achieved by applying cross-validation and by additional optimization of the training and model initialization process. To improve the classification results it is also possible to apply more advanced deep learning models than the basic CNN, as well as *cost-sensitive learning* approaches, in which a higher cost would be assigned to the incorrect classification of a subject with an affective disorder as healthy, given that such an error has more serious consequences than the incorrect classification of a healthy person. Finally, it would be interesting to investigate the combination of several different representations as input to the model, which could simultaneously exploit information about the signal in the time-frequency (spectrogram) and spatial (topographic maps) domains.

Although new methods of EEG signal classification and processing are being developed every day, there is still a series of approaches and directions for their further improvement that can contribute significantly to the improvement of diagnostic processes for mental disorders, including affective disorders. The results obtained within this thesis point to the potential for further development of systems based on deep learning in the field of EEG signal analysis, with a particularly emphasized need for expanding datasets, optimizing model architectures and developing approaches that additionally increase the clinical reliability and interpretability of the system.

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Statement on the use of artificial intelligence

Review of the use of artificial intelligence: During the writing of the thesis, ChatGPT (OpenAI, GPT-5) was occasionally used for language editing and rephrasing of individual sentences. All professional content, analysis of results and conclusions of the thesis are the result of the author's independent work.

Statement on the use of artificial intelligence: I confirm that during the writing of this final thesis I used artificial intelligence in accordance with the Policy on the Appropriate Use of Artificial Intelligence at the Faculty of Electrical Engineering and Computing, and that artificial intelligence did not replace my critical thinking or my contribution.